



Interstellar: An exploration into ray tracing within the influence of a black hole

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Plagiarism declaration

This piece of work is a result of my own work and I have complied with the Department's guidance on multiple submission and on the use of AI tools. Material from the work of others not involved in the project has been acknowledged, quotations and paraphrases suitably indicated, and all uses of AI tools have been declared.

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Declaration of use of AI tools in the writing process

Location	Explanation of AI use	AI used
Introduction	Utilised AI to suggest useful black hole history articles.	Gemini
Chapter 4 for light ray diagram	Used AI to create the diagram image in Latex using the Tikz module.	Gemini
Chapters 6	Utilised AI to help me understand links between special relativity and Euler-Largrange approaches to obtaining constants of motion. I could not find any articles which which explicitly do this so and wanted to ensure my understanding from mathematical physics was correct in this application.	Gemini
Chapters 5 and 7	Utilised the built in AI on visual studio code that helped speed up the programming process and debugging. As well as ask Gemini for module recommendations, such as Scipy and multiprocessing, to speed up processing.	Gemini, Copilot
Chapter 7	Utilised AI to help explain the concept of Bardeen equations and how to apply it in my code, required due to lack of resources on the topic.	Gemini

Abstract

A walk through on the mathematical derivations and concepts required to produce black hole visualisations like the one utilised in the movie Interstellar.

Ultimately, an exploration is taken into the specific coding method (DNGR code) utilised for the movie and a conclusion upon the accuracy of the black hole is made based upon the results derived in this paper along with expected properties we know from physics.

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Chapter 1

Introduction - Interstellar

In 1915 Albert Einstein proposed his theory of general relativity, this included the Einstein field equations which relate the curvature of spacetime with the mass, energy, and any momentum within it. In 1916, Karl Schwarzschild published his paper, [Schwarzschild \[2003\]](#), in which he utilised the Einstein field equations to produce the Schwarzschild solution, the first solution relating to a black hole (although at the time the concept of a black hole had not been established). Schwarzschild's solution described the spacetime around a static, spherically symmetric point mass which we now refer to as the Schwarzschild black hole. The actual concept of the black hole, however, was highly disregarded at the time, even by Einstein himself. In 1963 Roy Kerr discovered a metric for a rotating black hole, the solution for which is detailed in a publication from 1965, [Kerr and Schild \[2009\]](#). The Kerr metric is the metric for the Kerr black hole, the same black hole used in the movie Interstellar.

Interstellar is a famous movie released in 2014 that garnered 45 wins and 148 nominations to date according to IMDB. However, the significant accomplishment of Interstellar from a mathematical perspective is its stunning visualisation of a Kerr black hole. It was not until the 1960s that people (aware of its theory) began to accept the concept of black holes, and the actual visual interpretation of a black hole could only be left to the imagination of properties derived through mathematics. The first simulated image of a black hole was created in 1978 by a French physicist by the name of Jean-Pierre Luminet ([Lynn Gamwell](#)). Luminet managed to develop a black and white image of a black hole by hand-plotting the output he generated from a computer. The image depicted the shadow and distorted accretion disk of a black hole.

It was not until 2019 that the first real image of a black hole was taken by the Event Horizon Telescope. However, due to Interstellar, it was fair to say millions of people across the world had a visual conception of a spinning black hole before this historical event. The team behind Interstellar used mathematics and advanced code (DNGR code) to produce an accurate close up IMAX quality visualisation of a Kerr black hole for the movie, known as Gargantua. This dissertation delves into the mathematics utilised in the ray tracing required to produce a rendered visual like the black hole Gargantua, as well as the concepts required in the code itself.

Chapter 2

Spacetime

2.1 Understanding spacetime

To begin our journey into ray tracing around black holes, we will start by advancing our understanding of space as presented in general relativity.

It is important to know that in general relativity (which is the field of physics one must apply for such ray tracing) space is considered a 4 dimensional fabric. The four dimensions of such are time and three spacial co-ordinates.

A good visual way to interpret spacetime is **Figure 2.1**:

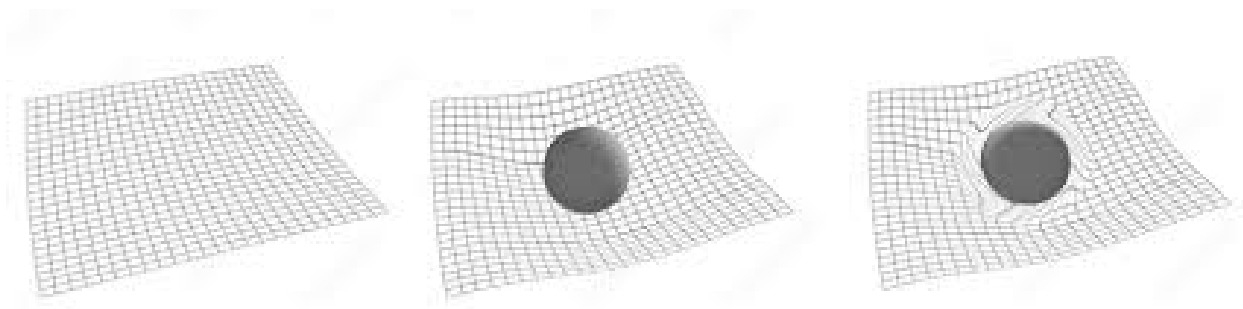


Figure 2.1: Spacetime

We can imagine space, without the time element, to be a simple mesh under a certain coordinate system, the basic coordinate system would be Cartesian. For the left most image we see an undisturbed grid of what could be (x,y,z) dimensional. In the other two images we see the effects of a mass being present in spacetime.

What Einstein taught us through general relativity (GR) is that gravity should not be considered a force as we were taught in Newtonian physics, but rather that gravity is the curvature of space time.

Intuitively, it has a rather simple visual component in that the greater the mass of a body in space, the more warped or disturbed space becomes. Thus, if there were a large body of mass warping space, the idea of the gravitational attraction of an object close by to that body is, rather, that it is falling into the "well" it has created as in **Figure 2.2**.

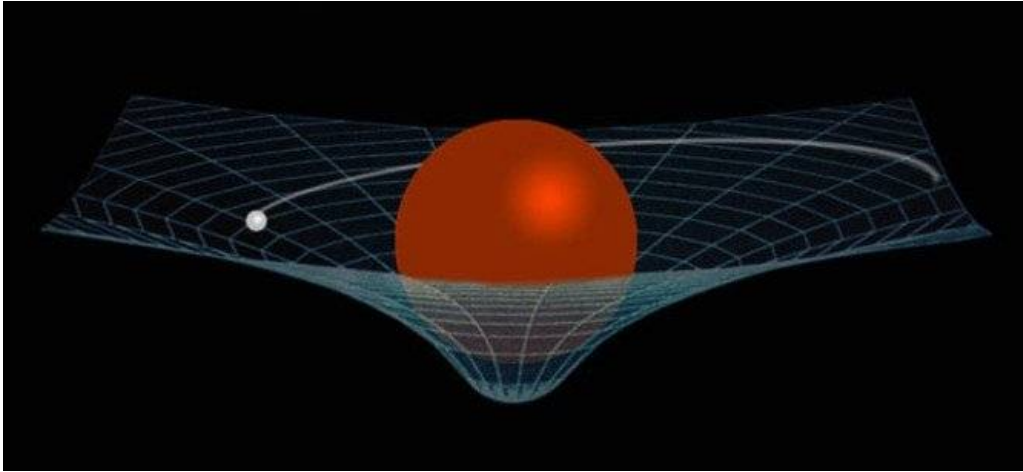


Figure 2.2: Orbits in GR

Now that we have an interpretation of the fabric of space we need to understand the full dimensional aspects in order to begin our mathematical journey.

2.2 Distances in spacetime

We discussed how spacetime is four-dimensional as we include the spacial and time components, but why? Why is it important to consider time in space?

Special relativity (SR) is the branch of relativity that unifies space and time whereas GR explains how spacetime distorts under mass. As such we will begin by referring to SR in this section but it will only briefly be touched upon for understanding purposes.

In SR we are taught that when measuring things in space we must consider time due to a relativistic property of movement. The whole concept of relativity comes from its name, we are considering relative motions. This means that when we measure things it is always in a relative frame. In our case of measuring trajectories a essential concept would be in measuring distances in space.

An example from the subsection **Spacetime in special relativity** from [Wikipedia \[g\]](#) is the spacetime interval example:

In three dimensions (Cartesian coordinates) we can measure the distance between two points as

$$(\Delta d)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2, \quad (2.1)$$

this being the Pythagorean theorem we are familiar with. Two observers can measure this distance (even using different coordinate systems, but assuming the same units) and get the same result, in SR we call this distance **invariant** (this will be explained in more detail later in the paper).

In SR however, the distance measured by the two observers will no longer be the same if one observer is moving relative to the other. This is due to relativistic affects such as **time dilation**

2.3 The spacetime interval

and **length contraction** which we will not delve into but will consider throughout this essay. If you wish to learn more about these concepts, the Wikipedia page provides greater detail.

"The fundamental reason for merging space and time into spacetime is that space and time are separately not invariant, which is to say that, under the proper conditions, different observers will disagree on the length of time between two events (because of time dilation) or the distance between the two events (because of length contraction)." -[Wikipedia \[g\]](#)

As such we introduce a new way of measuring distances in space whilst accounting for relativistic affects.

2.3 The spacetime interval

As in the previous section we discovered that the distance between two points in space are not invariant (measured to be the same in any frame) between two observers when moving relative to one another.

This has a negative consequence for us because when considering photon trajectories (light rays) we are considering the movement of such light rays relative to an observer, which we will consider stationary at an infinite distance away (this will become more clear later on when we begin solving for trajectories).

Thankfully there is a solution, the **spacetime interval**, also known as the **line element**. The spacetime interval is an invariant distance equation that takes into account the time component of spacetime. Being invariant means; no matter the observer, when taking into account the time and distance between two points, the same spacetime interval is calculated. The interval **in flat spacetime**, $(\Delta s)^2$, is calculated by

$$(\Delta s)^2 = (\Delta ct)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2. \quad (2.2)$$

The reason we define the interval as $(\Delta s)^2$ rather than Δs is due to the fact that intervals can be negative depending on the spacetime metric used (which is introduced in the next section). It is also important to note that depending on the **metric signature**, $(+ - - -)$ or $(- + + +)$, the interval can also be expressed as

$$(\Delta s)^2 = -(\Delta ct)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2. \quad (2.3)$$

So now we have an invariant interval in spacetime that combines distances in all space and time. We are a step closer to being able to derive equations of motion for light rays in spacetime. The next step is to introduce the notion of metrics.

2.4 Metric tensors

In the previous section we briefly mentioned the term **metric**. The metric tensor (the metric) is a vital component of spacetime. In the first section we discussed how the influence of gravity is actually the warping of spacetime (as visualised in 2.1).

When measuring spacetime intervals it is clear that if spacetime is warped, as though we were calculating regular three dimensional distances, we cannot use our conventional equation. The distance between two points in space changes entirely if space is warped along the path.

For this specific reason, Einstein introduced the metric tensor.

The metric tensor is essentially a tool that tells us how to convert distances according to the warping of spacetime in any coordinate system. A nice conceptual way of processing this is similar to the scale used on a map. Most maps will have a scale that tells you the true distance across a grid, maybe 5km per grid for instance, this is essentially the metric for that grid. Contours detailing elevation are another metric.

In the case of spacetime, however, the metric is a four dimensional tensor that details the factors in which we alter our spacetime interval.

Given a metric tensor for spacetime,

$$g_{\mu\nu} = \begin{bmatrix} g_{11} & g_{12} & g_{13} & g_{14} \\ g_{21} & g_{22} & g_{23} & g_{24} \\ g_{31} & g_{32} & g_{33} & g_{34} \\ g_{41} & g_{42} & g_{43} & g_{44} \end{bmatrix}. \quad (2.4)$$

We can express the components of $g_{\mu\nu}$ in terms of the basis vectors, $\vec{e}_\mu, \vec{e}_\nu$, of a given coordinate system by:

$$g_{\mu\nu} = \vec{e}_\mu \cdot \vec{e}_\nu. \quad (2.5)$$

It is important to note that the components are entirely dependent on the coordinate system and thus in warped space will usually be a function of the position due to a change in basis. Later in this chapter we will see how our spacetime interval in flat space (the Minkowski metric) changes under a new coordinate system.

2.4 Metric tensors

2.4.1 Examples of metric tensors

Let us look at the metric tensor for flat space, the Minkowski metric.

$$\eta_{\mu\nu} = g_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.6)$$

This is the metric we use for our equation (2.3). You can see how with a diagonal matrix like this the entries are merely the coefficients of the respective elements.

For a general metric tensor, $g_{\mu\nu}$, we have

$$(\Delta s)^2 = g_{\mu\nu} \begin{bmatrix} (dct)^2 \\ dx^2 \\ dy^2 \\ dz^2 \end{bmatrix} = \underbrace{g_{\mu\nu} dx^\mu}_{\text{Einstein notation}}. \quad (2.7)$$

Note: we will introduce Einstein notation in this chapter, but for now it is more important to familiarise the spacetime interval as a product of the metric tensor and coordinate vector for this example.

We will soon go on to look at one particular metric tensor which introduces us to the simplest of black holes, the Schwarzschild metric utilising spherical coordinates (r, θ, ϕ) :

$$g_{\mu\nu} = \begin{bmatrix} -(1 - \frac{2GM}{c^2 r}) & 0 & 0 & 0 \\ 0 & (1 - \frac{2GM}{c^2 r})^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}. \quad (2.8)$$

Where r is the radial position of a particle (or photon in our case), G is the gravitational constant, M is the mass of the black hole and θ, ϕ are spatial coordinates relative to the black hole.

2.5 Notation

It is important to note that in GR and SR we use Einstein notation. This is Einstein's summation notation taught in Analysis of Many Variables, however, in this case we use an index that is up called a **contravariant index** and an index that is down is called a **covariant index** to dictate summations. An important property is:

- when an index is expressed as a **Greek letter** it is used for time and space components, thus takes on value 0, 1, 2, or 3.
- If an index is expressed as a **Latin letter** it is used for spatial components only, thus takes on values 1, 2, or 3.

An example of this notation is

$$y = \sum_{i=1}^3 x^i e_i = x^1 e_1 + x^2 e_2 + x^3 e_3 \quad (2.9)$$

which is simplified to

$$y = x^i e_i \quad (2.10)$$

(This example was taken from [Wikipedia \[d\]](#), where further useful explanations and examples can be found).

2.6 Applying the metric

Now that we have established an understanding of our metric and the spacetime interval, given the fact that we will be measuring trajectories in curved space, it will be useful for us to find an expression of the spacetime interval in terms of the metric.

We can do this through the Equivalence principle, which states that at any point in spacetime there exists a local inertial frame where the laws of SR apply. This is a useful principle for us because in SR a local inertial frame refers to a locally flat space under the Minkowski metric (invariance holds).

Consider (2.3) in the Minkowski metric, i.e. flat space. We can write our generalised coordinates for flat space to be a function of the general coordinates (which could be curved or accelerating), x^α .

$$\zeta^\mu = \zeta^\mu(x^\alpha). \quad (2.11)$$

We can then apply multivariate chain rule for obtaining $d\zeta^\mu$

$$d\zeta^\mu = \frac{\partial \zeta^\mu}{\partial x^\alpha} dx^\alpha. \quad (2.12)$$

We previously expressed the spacetime interval as a summation of the metric tensor and coordinate vector (2.7). If we were to write out this summation for generalised coordinates

2.6 Applying the metric

under the minkowski metric it would look like

$$(ds)^2 = \eta_{\mu\nu} d\zeta^\mu d\zeta^\nu \quad (2.13)$$

$$= \eta_{\mu\nu} \frac{\partial \zeta^\mu}{\partial x^\alpha} dx^\alpha \frac{\partial \zeta^\nu}{\partial x^\beta} dx^\beta \quad (2.14)$$

$$= \left[\eta_{\mu\nu} \frac{\partial \zeta^\mu}{\partial x^\alpha} \frac{\partial \zeta^\nu}{\partial x^\beta} \right] dx^\alpha dx^\beta. \quad (2.15)$$

The expression $\eta_{\mu\nu} \frac{\partial \zeta^\mu}{\partial x^\alpha} \frac{\partial \zeta^\nu}{\partial x^\beta}$ defines the new metric.

$$g_{\alpha\beta} = \eta_{\mu\nu} \frac{\partial \zeta^\mu}{\partial x^\alpha} \frac{\partial \zeta^\nu}{\partial x^\beta}. \quad (2.16)$$

Intuitively, this expresses the components/coefficients of the change in coordinates under the new geometry.

Thus

$$(ds)^2 = g_{\alpha\beta} dx^\alpha dx^\beta. \quad (2.17)$$

Chapter 3

Geodesics

In general relativity Einstein made a clear statement: objects follow the straightest path through the curvature of spacetime. We call such paths **Geodesics**.

This means that for any metric we can find the path of a light ray between two points simply by finding the minimal path.

This allows us to derive a very useful equation called the **Geodesic equation**, which allows us to obtain equations of motion (EOM) for particles in space.

3.1 Deriving the general Geodesic equation

This derivation follows closely from the youtube video by [Faculty of Khan](#).

Let's denote the path of a particle under standard notation:

$$\gamma : \gamma(t) \text{ is a curve parametrised by } t, \gamma(t) = [x^0(t), x^1(t), x^2(t), x^3(t)]. \quad (3.1)$$

It is important to note that according to Einstein notation x^μ is the μ^{th} coordinate in spacetime rather than x to the power of μ .

Suppose γ exists in spacetime with the general metric tensor $g_{\mu\nu}$.

We can calculate the length of γ from $t = a$ to $t = b$ by:

$$l = \int_a^b \sqrt{|g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}|} dt. \quad (3.2)$$

We call this expression the action. It is important to note that this integrand is not the spacetime interval (2.3) but rather, it is the root of metric square of the four-velocity, u^μ , or rate of change of the path.

To get the length of a geodesic we require, as before mentioned, the minimum l for each coordinate x^μ . We can do this by utilising the technique of calculus of variations! In doing so we will be using the Euler-Lagrange equations.

We can write $(|g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}|)^{\frac{1}{2}}$ as a function of $t, x^0, \frac{dx^0}{dt}$ etc. $\implies F(t, x^0, \frac{dx^0}{dt}, \dots, x^3, \frac{dx^3}{dt})$.

Thus

$$l = \int_a^b (|g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}|)^{\frac{1}{2}} dt = \int_a^b F(t, x^0, \frac{dx^0}{dt}, \dots, x^3, \frac{dx^3}{dt}) dt. \quad (3.3)$$

3.1 Deriving the general Geodesic equation

To find the minimal length, l , we need solve the Euler-Lagrange equation for all coordinates x^α :

$$\frac{\partial F}{\partial x^\alpha} - \frac{d}{dt} \left(\frac{dF}{dx^{\alpha'}} \right) = 0, \quad \text{where } \alpha = 0, 1, 2, 3. \quad (3.4)$$

We can express F (our Lagrange equation) to be $F = \sqrt{(w)}$, where we will write w as being the metric square of the four velocity.

Plugging F into the Euler-Lagrange equation

$$\implies \frac{1}{2\sqrt{w}} \frac{\partial w}{\partial x^\alpha} - \frac{d}{dt} \left[\frac{1}{2\sqrt{w}} \frac{\partial w}{\partial x^{\alpha'}} \right] = 0. \quad (3.5)$$

If we assess the nature of $w = g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}$ we can simplify our derivatives:

$$\begin{aligned} g_{\mu\nu} &\leftarrow \text{function of } (x^0, x^1, x^2, x^3) \\ \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} &\leftarrow \text{function of } t \text{ only} \quad (3.1) \end{aligned}$$

Thus

$$\frac{\partial w}{\partial x^\alpha} = \frac{\partial g_{\mu\nu}}{\partial x^\alpha} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}. \quad (3.6)$$

When assessing $\frac{\partial w}{\partial x^{\alpha'}}$ it is important to note that whilst $g_{\mu\nu}$ is not a function of $x^{\alpha'}$ we cannot assume $\frac{\partial w}{\partial x^{\alpha'}} = 0$! We can show this by considering the summation of $g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}$:

$$\begin{aligned} g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} &= \sum_\mu \sum_\nu g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \\ &= g_{00} \frac{dx^0}{dt} \frac{dx^0}{dt} + \dots + g_{\mu\alpha} \frac{dx^\mu}{dt} \frac{dx^\alpha}{dt} + \dots + g_{\alpha\mu} \frac{dx^\alpha}{dt} \frac{dx^\mu}{dt} + \dots \\ &= g_{00} (x^{0'})^2 + \dots + 2g_{\mu\alpha} x^{\alpha'} x^{\mu'} + \dots \end{aligned}$$

Here we can see that w contains the term $x^{\alpha'}$. **Note**, we used the fact that metric tensors are symmetric to deduce $g_{\mu\alpha} = g_{\alpha\mu}$.

Thus

$$\frac{\partial w}{\partial x^{\alpha'}} = 2g_{\mu\alpha} \frac{dx^\mu}{dt}. \quad (3.7)$$

Using (3.6) and (3.7), we can return to the Euler-Lagrange equations and find

$$\begin{aligned}
 \frac{1}{2\sqrt{w}} \frac{\partial g_{\mu\nu}}{\partial x^\alpha} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} &= \frac{d}{dt} \left[\frac{1}{\sqrt{w}} g_{\mu\alpha} \frac{dx^\mu}{dt} \right] \\
 \text{(RHS product rule)} &= -\frac{1}{2w\sqrt{w}} \frac{dw}{dt} g_{\mu\alpha} \frac{dx^\mu}{dt} + \frac{1}{\sqrt{w}} \frac{dg_{\mu\alpha}}{dt} \frac{dx^\mu}{dt} + \frac{1}{\sqrt{w}} g_{\mu\alpha} \frac{d^2 x^\mu}{dt^2} \\
 \text{(product rule on } g_{\mu\alpha}) &= -\frac{1}{2w\sqrt{w}} \frac{dw}{dt} g_{\mu\alpha} \frac{dx^\mu}{dt} + \frac{1}{\sqrt{w}} \frac{\partial g_{\mu\alpha}}{\partial x^\nu} \frac{dx^\nu}{dt} \frac{dx^\mu}{dt} + \frac{1}{\sqrt{w}} g_{\mu\alpha} \frac{d^2 x^\mu}{dt^2}.
 \end{aligned}$$

Note: by chain rule

$$\frac{dg_{\mu\alpha}}{dt} = \frac{\partial g_{\mu\alpha}}{\partial x^0} \frac{dx^0}{dt} + \frac{\partial g_{\mu\alpha}}{\partial x^1} \frac{dx^1}{dt} + \dots + \frac{\partial g_{\mu\alpha}}{\partial x^\nu} \frac{dx^\nu}{dt} = \frac{\partial g_{\mu\alpha}}{\partial x^\nu} \frac{dx^\nu}{dt}.$$

Where ν is a dummy index.

Multiplying the full equation by $2\sqrt{w}$ and rearranging we get

$$\begin{aligned}
 \frac{1}{w} \frac{dw}{dt} g_{\mu\alpha} \frac{dx^\mu}{dt} &= 2 \frac{\partial g_{\mu\alpha}}{\partial x^\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} + 2g_{\mu\alpha} \frac{d^2 x^\mu}{dt^2} - \frac{\partial g_{\mu\nu}}{\partial x^\alpha} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \\
 &= \underbrace{\left(\frac{\partial g_{\mu\alpha}}{\partial x^\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} + \frac{\partial g_{\mu\alpha}}{\partial x^\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \right)}_{\substack{\mu \text{ and } \nu \text{ are dummy variables so can swap } \mu \text{ and } \nu}} + 2g_{\mu\alpha} \frac{d^2 x^\mu}{dt^2} - \frac{\partial g_{\mu\nu}}{\partial x^\alpha} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \\
 \implies \frac{1}{w} \frac{dw}{dt} g_{\mu\alpha} \frac{dx^\mu}{dt} &= \frac{\partial g_{\mu\alpha}}{\partial x^\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} + \frac{\partial g_{\alpha\mu}}{\partial x^\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} + 2g_{\mu\nu} \frac{d^2 x^\mu}{dt^2} - \frac{\partial g_{\mu\nu}}{\partial x^\alpha} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \\
 &= \underbrace{\left(\frac{\partial g_{\mu\alpha}}{\partial x^\mu} + \frac{\partial g_{\alpha\mu}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\alpha} \right)}_{2\Gamma_{\mu\nu\alpha} \text{ by definition}} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} + 2g_{\mu\alpha} \frac{d^2 x^\mu}{dt^2} \\
 \implies \frac{1}{w} \frac{dw}{dt} g_{\mu\alpha} \frac{dx^\mu}{dt} &= 2\Gamma_{\mu\nu\alpha} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} + 2g_{\mu\alpha} \frac{d^2 x^\mu}{dt^2}.
 \end{aligned}$$

Where $\Gamma_{\mu\nu\alpha}$ is the **Christoffel symbol**. See (3.1.1).

If we multiply our geodesic equation by the inverse metric tensor components, $g^{\sigma\alpha}$, where σ is the chosen coordinate space, we get:

$$\implies 2 \underbrace{g^{\sigma\alpha} g_{\mu\alpha}}_{\delta_\mu^\sigma} \frac{d^2 x^\mu}{dt^2} + 2 \underbrace{g^{\sigma\alpha} \Gamma_{\mu\nu\alpha}}_{\Gamma_{\mu\nu}^\sigma} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} = \frac{1}{w} \frac{dw}{dt} \underbrace{g^{\sigma\alpha} g_{\mu\alpha}}_{\delta_\mu^\sigma} \frac{dx^\mu}{dt}$$

3.1 Deriving the general Geodesic equation

Thus our equation becomes,

$$\frac{d^2 x^\sigma}{dt^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} = \frac{1}{2w} \frac{dw}{dt} \frac{dx^\sigma}{dt}. \quad (3.8)$$

We can further simplify this equation by considering the arc length, s , of the geodesic. This will give us our general path geodesic.

We again consider the expression $g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}$, mathematically the metric square of the four-velocity is the rate of change of the path squared, as such we can express l as

$$l = \int_a^b \underbrace{\sqrt{g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}}}_{\frac{ds}{dt}} dt = \int_A^B ds. \quad (3.9)$$

w can be written as $\left(\frac{ds}{dt}\right)^2$, but if t is the arc length parameter s , then $w = \left(\frac{ds}{dt}\right)^2 = 1$. Thus, $\frac{dw}{dt} = \frac{dw}{ds} = 0$.

This leaves us with:

$$\frac{d^2 x^\sigma}{ds^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{ds} \frac{dx^\nu}{ds} = 0. \quad (3.10)$$

In special relativity we are taught that proper time, τ , can be related to a particles path by

$$\tau = \frac{\sqrt{-s^2}}{c} \rightarrow d\tau = \frac{\sqrt{-(ds)^2}}{c}.$$

Thus we can obtain the geodesic equation in terms of τ rather than s (we would use this equation to find EOM of any particle with mass),

$$\frac{d^2 x^\sigma}{d\tau^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0. \quad (3.11)$$

How can we apply the geodesic equation to light rays?

The issue with light rays is that due to the photon travelling at the speed c , we find the distance covered to be proportional to the time elapsed ($ds^2 = -c^2 dt^2 + \underbrace{dx^2 + dy^2 + dz^2}_{=c^2 dt^2} = 0$) (2.3).

We have linked proper time to the spacetime interval and thus we know that differentiating with respect to proper time for a light ray would be counter-intuitive. To tackle this problem we introduce an **affine parameter**, λ .

The function of the affine parameter is to parametrise the path of the photon along the geodesic. The parameter itself can be anything that preserves the geodesic equation hence the term "affine", [Wikipedia \[a\]](#). The Geodesic we get under λ is

$$\Rightarrow \frac{d^2 x^\sigma}{d\lambda^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0. \quad (3.12)$$

3.1.1 Christoffel symbols

Above we introduced the notation of the Christoffel symbol, $\Gamma_{\sigma\mu\nu}$, which in terms of our general geometric space looks like:

$$\Gamma_{\sigma\mu\nu} = \frac{1}{2} \left(\frac{\partial g_{\sigma\mu}}{\partial x^\nu} + \frac{\partial g_{\sigma\nu}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\sigma} \right)$$

$$\Rightarrow \Gamma_{\mu\nu}^\sigma = \frac{1}{2} g^{\sigma\beta} \left(\frac{\partial g_{\beta\mu}}{\partial x^\nu} + \frac{\partial g_{\beta\nu}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\beta} \right). \quad (\beta \text{ is the dummy index we sum over here})$$

If we were to assess this equation it becomes clear to us that the Christoffel symbol is an expression of the change in spatial geometry given a change in coordinates. "In words, the arrays represented by the Christoffel symbols track how the basis changes from point to point" - [Wikipedia \[b\]](#).

This becomes more apparent when we remind ourselves of the definition of the metric as the dot product of the respective basis vectors, which we defined at the beginning of chapter 2 (2.7).

A good example would be, in the euclidean space the Christoffel symbol becomes the simple expression:

$$\Gamma_{\sigma\mu\nu} = \frac{\partial \mathbf{e}_\nu}{\partial x^\mu} \cdot \mathbf{e}_\sigma,$$

$$\Gamma_{\mu\nu}^\sigma = \frac{\partial \mathbf{e}_\nu}{\partial x^\mu} \cdot g^{\sigma\beta} \mathbf{e}_\beta. \quad (3.13)$$

3.1.2 Example case: flat space

To solidify our understanding of the geodesic equation we will take a look at a simple example from [David Tong](#).

Consider the flat space in $\mathbb{R}^3$ with polar coordinate system:

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$\Rightarrow (ds)^2 = (dr)^2 + r^2(d\theta)^2 + r^2 \sin^2 \theta (d\phi)^2.$$

The metric is

$$g_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}.$$

Note : σ runs over the coordinate system (r, θ, ϕ) .

We can now use the geodesic equation to obtain the EOM for a straight line in flat space.

First we calculate the Christoffel symbols using

$$\Gamma_{ij}^\sigma(x) = \frac{1}{2} g^{\sigma l} \left(\frac{\partial g_{li}}{\partial x^j} + \frac{\partial g_{lj}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right). \quad (3.14)$$

3.1 Deriving the general Geodesic equation

For $\Gamma_{\theta\theta}^r$ we calculate

$$\Gamma_{\theta\theta}^r = \frac{1}{2} \underbrace{g^{rl}}_{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}} (2\partial_\theta \underbrace{g_{l\theta}}_{\begin{pmatrix} 0 \\ 0 \\ r^2 \sin^2 \theta \end{pmatrix}} - \underbrace{\partial_l g_{\theta\theta}}_{\begin{pmatrix} dr \\ d_\theta \\ d_\theta \end{pmatrix}}) \quad (3.15)$$

$$= \frac{1}{2}(-\partial_r(r^2)) = -r. \quad (3.16)$$

It is important to note that g^{ij} is the **inverse matrix** of g_{ij} such that

$$g^{ij} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^{-2} & 0 \\ 0 & 0 & (r^2 \sin^2 \theta)^{-1} \end{bmatrix}. \quad (3.17)$$

The inverse is easy to calculate here due to the matrix's diagonal property.

We can calculate the rest of the Christoffel symbols to get

$$\begin{aligned} \implies \Gamma_{\theta\theta}^r &= -r, & \Gamma_{\phi\phi}^r &= -r \sin^2 \theta, & \Gamma_{\theta r}^\theta &= \Gamma_{r\theta}^\theta = \frac{1}{r} \\ \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta, & \Gamma_{\phi r}^\phi &= \Gamma_{r\phi}^\phi = \frac{1}{r}, & \Gamma_{\theta\phi}^\phi &= \Gamma_{\phi\theta}^\phi = \frac{\cos \theta}{\sin \theta} \end{aligned}$$

We can now plug these Christoffel symbols into (3.11) and rearrange to get our equations of motion for geodesics in flat space:

$$\begin{aligned} \ddot{r} &= r\dot{\theta}^2 + r \sin^2 \theta \dot{\phi}^2, \\ \ddot{\theta} &= \sin \theta \cos \theta \dot{\phi}^2 - \frac{2}{r} \dot{\theta} \dot{r}, \\ \ddot{\phi} &= -\left(\frac{2}{r} \dot{\theta} \dot{r} + 2 \frac{\cos \theta}{\sin \theta} \dot{\theta} \dot{\phi} \right). \end{aligned}$$

Which simplifies to:

$$\begin{aligned} \ddot{r} &= r\dot{\theta}^2 + r \sin^2 \theta \dot{\phi}^2, \\ \frac{d}{dt}(r^2 \dot{\theta}) &= r^2 \sin \theta \cos \theta \dot{\phi}^2, \\ \frac{d}{dt}(r^2 \sin^2 \theta \dot{\phi}) &= 0. \end{aligned}$$

These results align with what we would expect to obtain if we converted the Cartesian EOM, $\ddot{x}$, $\ddot{y}$, $\ddot{z}$ into polar coordinates.

Chapter 4

The Schwarzschild metric

So far we have evolved our understanding of spacetime, this led us to introducing metric tensors which we then used to define our general geodesic equation. Ultimately, this chapter will progress to solve the geodesic equations in a given metric, the Schwarzschild metric, to find the given light ray trajectories.

We previously introduced the Schwarzschild metric in chapter 2 as an example, but now we will see the true significance of this metric.

4.1 What is the Schwarzschild metric?

We previously introduced the metric by the metric tensor that applies to the simplest black hole, the Schwarzschild black hole. The tensor being

$$g_{\mu\nu} = \begin{bmatrix} -(1 - \frac{2GM}{c^2 r}) & 0 & 0 & 0 \\ 0 & (1 - \frac{2GM}{c^2 r})^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}. \quad (4.1)$$

Where r is the radial position of a particle (or photon in our case), G is the gravitational constant, M is the mass of the black hole and θ, ϕ are spatial coordinates relative to the black hole. **Note:** the Schwarzschild metric has a metric signature of $(-+++)$ and utilises the spherical coordinate system.

What exactly makes the Schwarzschild black hole simple? The answer is that the Schwarzschild metric's spacetime is assumed to be **spherically symmetrical** and **static**. The importance of these properties is that it allows us to significantly simplify the equations we will utilise. Here is how:

4.1.1 Symmetry

When considering a body of mass in space we can only make calculations of motion for an object nearby that body based on two conditions, solving for a component of velocity or solving for the acceleration. When solving for either of the two we are considering a four dimensional problem which can lead to significantly large equations. By considering plane symmetry in spacetime we are able to reduce the number of dimensions required in these equations of motion we come to derive.

This is why the Schwarzschild metric is considered the foundational metric of black holes; it introduces us to clear and simple equations, by reducing dimensions involved, representing light ray motion before we move on to the more complex Kerr metric.

4.2 The origins of the Schwarzschild metric

4.1.2 Static

A static spacetime is one in which all metric components are independent of the time coordinate, τ , (so that $\frac{\partial}{\partial \tau} g_{\mu\nu} = 0$). This holds significant meaning in our geodesic equations which we will solve under this metric later in this chapter.

In being static spacetime, we are assuming that the black hole is non-spinning which has significant influence on our light ray trajectories.

4.2 The origins of the Schwarzschild metric

An important question is how are these metrics derived? The simple answer is through the **Einstein field equations** and the assumptions of the spacetime which the metric represents. Deriving the metric is not important for us but is an interesting topic which can be found on [Wikipedia \[c\]](#).

It was through the derivation of the metric that Karl Schwarzschild derived the **Schwarzschild radius**, r_s . However, at the time he did not explicitly define this constant and merely interpreted it to be a feature of the radial coordinate for his point mass solution to Einstein's field equations (as found in his paper [Schwarzschild \[2003\]](#) where it states α being a constant depending on the mass). The Schwarzschild radius is very significant as it is the radius at which light cannot escape the well of gravity a significant mass produces, also known as the radius of the event horizon. All black holes have an event horizon and it is a significant property because we will use this knowledge to impose constraints for finding light ray trajectories when we look at ray tracing.

4.3 Solving for light ray trajectories in the Schwarzschild metric

This part of the chapter follows from a brilliantly explained YouTube video by [eigenchris](#).

4.3.1 Solving the geodesic equation

We obtained the geodesic equation in Chapter 3 ([3.12](#)) under the affine parameter, λ .

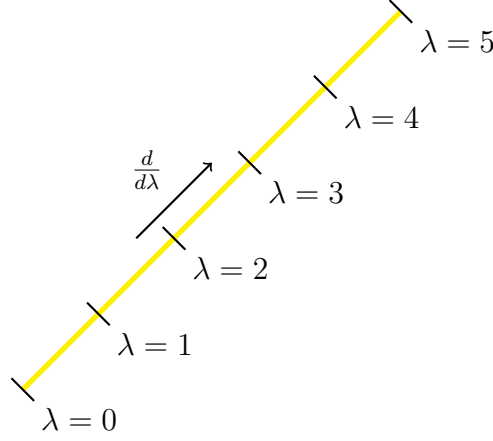
$$\frac{d^2 x^\sigma}{d\lambda^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0.$$

We will solve this equation to obtain some useful constants of motion as well as obtain the ordinary differential equations (ODEs) that represent the motion of light rays.

Light rays are massless and we previously discussed how proper time, τ , and the spacetime interval, ds^2 , for a photon are equal to 0 in section [3.1](#).

We established that the purpose of λ is that it parametrises the geodesic. To be more precise, it equally segments the path with its tangent vector, $\frac{d}{d\lambda}$, constant and along the light ray's world line (the path taken in spacetime). Thus, $\frac{d}{d\lambda} \cdot \frac{d}{d\lambda} = \left\| \frac{d}{d\lambda} \right\|^2 = 0$ and we call these vectors light-like/null. Geodesics for light rays are correspondingly called **null geodesics**.

4.3 Solving for light ray trajectories in the Schwarzschild metric



By using λ instead of τ our geodesic equation works for both massive and massless particles.

This geodesic equation allows us to produce four separate equations, one for each spacetime coordinate:

$$\frac{d^2 x^\sigma}{d\lambda^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0.$$

$$0 = \frac{d^2(ct)}{d\lambda^2} + \Gamma_{\mu\nu}^t \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}, \quad (4.2)$$

$$0 = \frac{d^2\phi}{d\lambda^2} + \Gamma_{\mu\nu}^\phi \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}, \quad (4.3)$$

$$0 = \frac{d^2 r}{d\lambda^2} + \Gamma_{\mu\nu}^r \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}, \quad (4.4)$$

$$0 = \frac{d^2\theta}{d\lambda^2} + \Gamma_{\mu\nu}^\theta \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}. \quad (4.5)$$

We can calculate the Christoffel symbols for these equations to be:

$$\Gamma_{tr}^t = \Gamma_{rt}^t = \frac{r_s}{2r(r-r_s)}, \quad \Gamma_{tt}^r = \frac{r_s(r-r_s)}{2r^3}, \quad \Gamma_{rr}^r = \frac{r_s}{2r(r_s-r)}, \quad \Gamma_{\theta\theta}^r = r_s - r,$$

$$\Gamma_{\phi\phi}^r = (r_s - r) \sin^2 \theta, \quad \Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \Gamma_{r\phi}^\phi = \Gamma_{\phi r}^\phi = \frac{1}{r},$$

$$\Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta, \quad \Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi = \cot \theta.$$

We will look at (4.2) and (4.3) to derive some useful constants of motion.

Note: before we established that the symmetric property of the Schwarzschild metric allows us to simplify our equations, we directly apply this going forward by taking the orbit to be on the plane of the equator. This means we can **fix** $\theta = \frac{\pi}{2}$, and thus eliminates the need to consider (4.5).

(4.2):

$$\begin{aligned} 0 &= \frac{d^2(ct)}{d\lambda^2} + \Gamma_{\mu\nu}^t \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = \frac{d^2(ct)}{d\lambda^2} + 2\Gamma_{tr}^t \frac{d(ct)}{d\lambda} \frac{dr}{d\lambda} \\ &= \frac{d^2(ct)}{d\lambda^2} + \frac{r_s}{r(r-r_s)} \frac{d(ct)}{d\lambda} \frac{dr}{d\lambda}. \end{aligned}$$

4.3 Solving for light ray trajectories in the Schwarzschild metric

Applying chain rule to this equation we obtain

$$0 = \frac{d}{d\lambda} \left(\frac{d(ct)}{d\lambda} \left(1 - \frac{r_s}{r} \right) \right) \implies \frac{d(ct)}{d\lambda} \left(1 - \frac{r_s}{r} \right) = \epsilon. \quad (\epsilon \text{ being a constant of motion})$$

We denote $\epsilon = \frac{E}{m}$ as the energy per unit mass which can be derived through taking the limit $r \rightarrow \infty$:

$$\frac{d(ct)}{d\lambda} \left(1 - \frac{r_s}{r} \right) \rightarrow c \frac{dt}{d\lambda} = \frac{E}{m} \implies E = mc\gamma.$$

This expression for energy is equivalent to p^0 in SR, that being the first element of the 4-momentum of a particle, where $\frac{dt}{d\lambda} = \gamma$ is the Lorentz factor.

(4.3):

$$\begin{aligned} 0 &= \frac{d^2 \phi}{d\lambda^2} + \Gamma_{\mu\nu}^{\phi} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = \frac{d^2 \phi}{d\lambda^2} + 2\Gamma_{r\phi}^{\phi} \frac{dr}{d\lambda} \frac{d\phi}{d\lambda} + 2\Gamma_{\theta\phi}^{\phi} \frac{d\theta}{d\lambda} \frac{d\phi}{d\lambda} \\ &= \frac{d^2 \phi}{d\lambda^2} + \frac{2}{r} \frac{dr}{d\lambda} \frac{d\phi}{d\lambda} + 2 \cot \frac{d\theta}{d\lambda} \frac{d\phi}{d\lambda} = \frac{d^2 \phi}{d\lambda^2} + \frac{2}{r} \frac{dr}{d\lambda} \frac{d\phi}{d\lambda} \quad (\text{Recall: } \theta = \frac{\pi}{2}). \end{aligned}$$

By chain rule we once again derive a constant of motion:

$$\frac{d^2 \phi}{d\lambda^2} + \frac{2}{r} \frac{dr}{d\lambda} \frac{d\phi}{d\lambda} = \frac{d}{d\lambda} \underbrace{\left(r^2 \frac{d^2 \phi}{d\lambda^2} \right)}_{\text{C.O.M}} = 0$$

We denote this as the Angular momentum per unit mass, l .

$$l = \frac{L}{m} = r^2 \frac{d\phi}{d\lambda}. \quad (4.6)$$

These constants of motion can be utilised to simplify our equations of motion into simple ODEs. Not only are they useful for simplifying our equations, but they can also be used to derive a physical property of black holes, the **photon sphere**.

Let us first derive the photon sphere before we move onto our equations of motion.

Utilising the metric square of the photon's four velocity, which we saw in chapter 3, with respect to the affine parameter, λ :

$$\begin{aligned} g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} &= \left(\frac{d(ct)}{d\lambda} \right)^2 g_{tt} + \left(\frac{dr}{d\lambda} \right)^2 g_{rr} + \left(\frac{d\theta}{d\lambda} \right)^2 g_{\theta\theta} + \left(\frac{d\phi}{d\lambda} \right)^2 g_{\phi\phi} \\ &= \left(\frac{d(ct)}{d\lambda} \right)^2 \left(1 - \frac{r_s}{r} \right) - \left(\frac{dr}{d\lambda} \right)^2 \left(1 - \frac{r_s}{r} \right)^{-1} - \left(\frac{d\phi}{d\lambda} \right)^2 r^2. \end{aligned}$$

Substituting our constants of motion into this equation and accounting for the fact that, mathematically,

$$g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = \frac{ds^2}{d\lambda^2} = 0. \quad (ds^2 = 0)$$

$$\epsilon^2 \left(\frac{r}{r - r_s} \right)^2 \left(1 - \frac{r_s}{r} \right) - \left(\frac{dr}{d\lambda} \right)^2 \left(1 - \frac{r_s}{r} \right)^{-1} - \left(\frac{l}{r^2} \right)^2 = 0.$$

$$\implies 0 = \epsilon^2 - \left(\frac{dr}{d\lambda} \right)^2 - \left(1 - \frac{r_s}{r} \right) \frac{l^2}{r^2}$$

From this equation we can obtain the expression for ϵ (energy per unit mass) as:

$$\epsilon^2 = \underbrace{\left(\frac{dr}{d\lambda} \right)^2}_{K.E} + \underbrace{\frac{l^2}{r^2} - \frac{l^2 r_s}{r^3}}_{V_{eff}(r)} \quad (4.7)$$

$K.E$ being kinetic energy and $V_{eff}(r)$ the effective potential of our photons.

Using this expression we can obtain the radius at which light has a fixed orbit, also known as, the **photon sphere** of a Schwarzschild black hole.

We set $\frac{dr}{d\lambda} = 0$:

$$\begin{aligned} \epsilon^2 &= \frac{l^2}{r^2} - \frac{l^2 r_s}{r^3} \implies \frac{d}{dr} (V_{eff}(r)) = 0 \\ 0 &= \frac{d}{dr} \left(\frac{l^2}{r^2} - \frac{l^2 r_s}{r^3} \right) \\ 0 &= -2 \frac{l^2}{r^3} + 3 \frac{l^2 r_s}{r^4} \implies -2l^2 r + 3l^2 r_s \\ \implies r &= \frac{3}{2} r_s = 3M \frac{G}{c^2}. \end{aligned}$$

Later we will see demonstrations of this feature in our visualisations.

From the Geodesic equations and constants of motion (COM), we set $G = c = 1$ to obtain the EOMs:

$$\dot{\phi} = \frac{l}{r^2}, \quad (4.8)$$

$$\dot{r} = s, \quad (4.9)$$

$$\dot{s} = -\frac{l^2(3M - r)}{r^4}. \quad (4.10)$$

We will go on to utilise these in ray tracing under the Schwarzschild metric.

Chapter 5

Ray Tracing under the Schwarzschild metric

Utilising the EOMs from chapter 4, we can begin the ray tracing process for the Schwarzschild metric.

5.1 Modelling light ray trajectories

We begin with a 2D plot which we develop in Python. We have a black hole positioned at $r = 0$, and a camera facing towards it from a near infinite distance, r_{camera} (both lie on the x-axis for our 2D plot and data). We call the surrounding space of these objects the celestial sphere. The camera being on the equatorial plane ($\theta = \frac{\pi}{2}$) allows us to reduce the involved dimensions from a spherical to polar coordinate system.

We apply the concept of **backwards integration** to the light rays, in which we integrate backwards in time from a point on the camera lens back to the source of a ray. A useful function in Python that allows to easily solve our ODEs is the `Solve_ivp` function in the Scipy module. This function allows us to add conditional events to our integrations which we will specify to be the capture of a ray or the escape of a ray.

Having established a field of view (FOV) for our camera, which in our case will be 90 degrees, we sample our incoming rays over the angle interval $\beta \in [-45^\circ, 45^\circ]$. To begin our code we must establish initial conditions.

We initialise the angular momentum, l , hitting the camera (as the camera is near infinite distance away we consider the initial momentum to adhere to flat spacetime). This can be obtained by the radius multiplied by linearised momentum, $l = r \cdot p \sin \beta = r_{\text{camera}} \cdot \frac{E}{c} \sin \beta$. In our code we set $E = c = 1$ for simple scaling. We obtain expressions for potential energy, V^2 , and kinetic energy, $\dot{r}^2$, from (4.7):

$$l = r_{\text{camera}} \sin(\beta), \quad (\text{this being the initial tangential momentum}) \quad (5.1)$$

$$V^2 = \left(1 - \frac{2M}{r_{\text{camera}}}\right) \frac{l^2}{r^2}, \quad (\text{the potential energy squared}) \quad (5.2)$$

$$\dot{r}^2 = \epsilon^2 - V^2. \quad (5.3)$$

Also using $\epsilon = G = 1$, fixed M and r_{camera} . From these initial conditions and our EOMs, (4.8), (4.9) and (4.10) we can develop a right hand side condition for our `Solve_ivp` function. We apply our conditional events that will omit the integration of rays that are captured or continue the integration of escaped rays up until they hit the celestial sphere. We apply this function to

our range of rays with incoming angles, β , to the camera and we yield the following results:

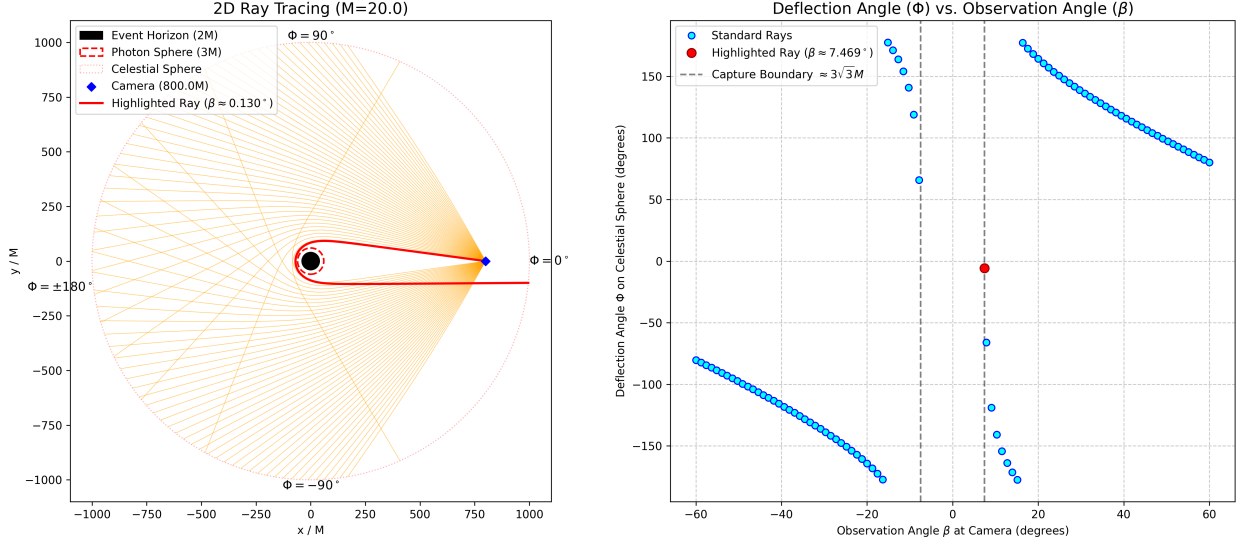


Figure 5.1: Plots with $r_{\text{camera}} = 800, r_{\text{sphere}} = 1000$

Here we can have a visual understanding of gravitational lensing around a Schwarzschild black hole. The left plot in figure 5.1 above shows the ray trajectories, including a highlighted example ray. The right plot is a scatter graph of the angle of the ray's source point (we call the deflection angle) on the celestial sphere, $\phi \in [-180^\circ, 180^\circ]$ against the incoming angle of the ray to the lens, $\beta \in [-45^\circ, 45^\circ]$.

The right plot demonstrates the asymptotic behaviour of lensing around the black hole, where we have included the capture boundary at $r = 3\sqrt{3}M$. This capture boundary radius lies just beyond the photon sphere we previously derived in chapter 4 and represents the point where rays begin to fall into the black hole. Our highlighted ray path follows close to this boundary.

5.2 What would we expect to see?

Thus far we have established the event horizon of a Schwarzschild black hole; we have derived the photon sphere for a Schwarzschild black hole; however, we only discussed them briefly. Perhaps the reader has never seen a visualisation of black hole, or perhaps they have seen multiple different images for black holes and are unable to discern their features. This section will make it clear what it is we would expect to see from a Schwarzschild black hole image.

We discussed that the event horizon of the black hole is the point at which light cannot escape the well of spacetime, this solidifies that concept of the black hole shadow. All black holes will have a shadow due to the absence of light rays that we would expect to come from that area of space. Black hole shadows vary in size depending on their mass and their shape can vary depending on the type of black hole, we will cover this when we discuss Kerr black holes, but for a Schwarzschild black hole we find the shadow to be circular. Under this metric, we discussed how the photon sphere is the point at which light orbits the black hole. We find that the photon sphere lies quite close to the event horizon (relative to the black hole's mass) and

5.3 Lensing visualisation

visually we do not see the photon sphere but we do see light rays whose trajectories lie close to or even within the photon sphere. It is important to note that the photon sphere is a region of instability and as such not all light rays permanently orbit the black hole around this radius so we can see a circular ring of light due to these rays that encapsulate the shadow.

Beyond these two main features, the most prominent thing we would see is **gravitational lensing**. "A gravitational lens is matter, such as a cluster of galaxies or a point particle, that bends light from a distant source as it travels toward an observer." - [Wikipedia \[e\]](#). The Schwarzschild black hole is the body of mass that lenses light rays for the observer. Visually, we would view lensing as a warping, stretching and even translation of light sources (particularly stars) in space as we have demonstrated above with our 2D plots. The biggest lensing feature is known as the Einstein ring. Light rays that emerge from a source on the opposite side of the black hole, and aligned with the observer, forms a ring of light (the Einstein ring) around the shadow that we see due to maximum distortion where the light deflects around the black hole. Lensing will become more apparent when we generate our own lensed image in this chapter. For this paper we only require a basic understanding of gravitational lensing, however a detailed explanation can be found in the article [James et al. \[2015\]](#).

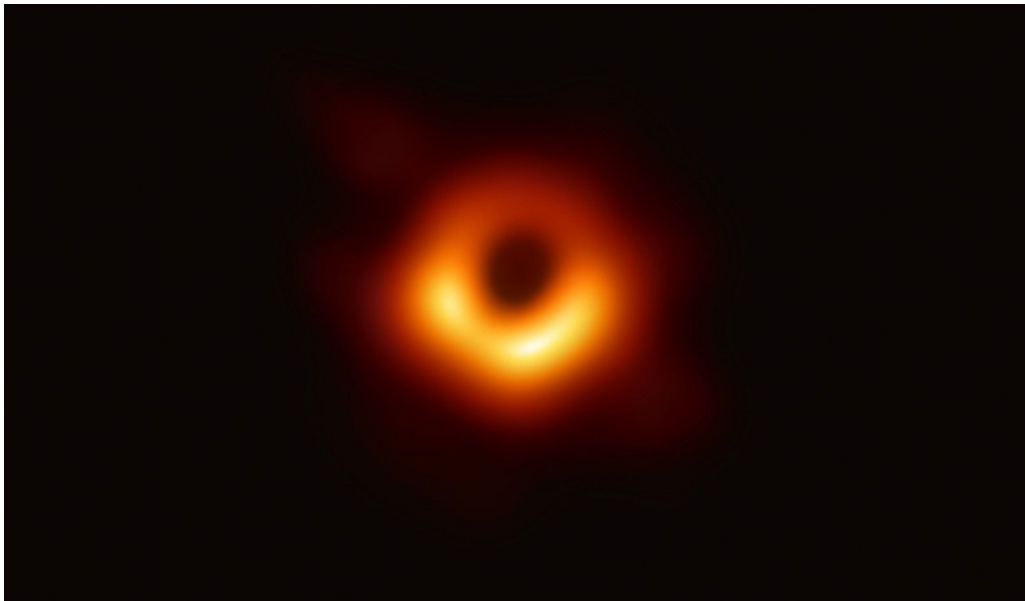


Figure 5.2: First black hole (Kerr black hole) image demonstrating the shadow and photon sphere. Captured by Nasa, 2019.

5.3 Lensing visualisation

Now that we have laid out the foundation for ray tracing and established a 2 dimensional data frame, we can begin to generate better visuals. We utilise our generated data for β in order to manipulate an image to represent gravitational lensing. The goal is to map each ray from a certain point on the camera lens to a pixel on the image; in doing so we will translate pixels in which represent source points for the rays to their lensed points.

We begin by "unfolding" our celestial sphere into a 1920 x 1080p image (which we will call our celestial map) and convert it into an array of pixels, using Python's PIL module, which is

essentially just a grid of pixels and their corresponding RGB values. We form another array for the camera lens, this will be a grid spanning 1920 points on the x axis and 1080 points on the y axis.

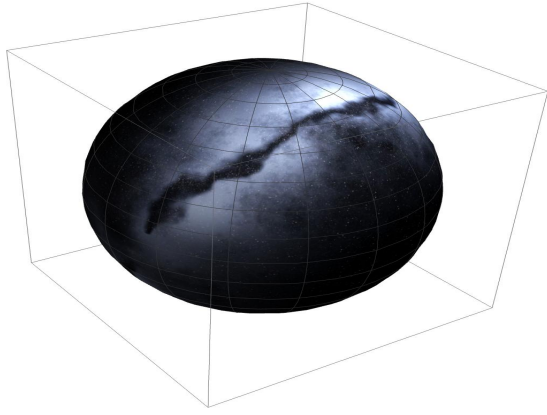
We consider a point (x, y) on our screen (1920 x 1080 array) in front of the camera at a focal lengths distance, f (this is a distance determined by the cameras FOV and screen width), away from our considered lens. We now follow the process:

- Each pixel on the screen is assigned a vector, $V = [f, x, y]$.
- We consider the x-axis as the 3D vector $u = [1, 0, 0]$ and calculate the angle between V and u , $\beta = \arccos\left(\frac{u \cdot V}{|V|}\right)$. This β corresponds to our incoming angle.
- We interpolate for any missing β and obtain the corresponding ϕ for each ray.
- Spherical symmetry means the ray remains in a single plane, n . We find n by taking the normal vector of both u and V , $n = u \times V$, this plane is the plane on which the ray rotates.
- We calculate a final 3D vector $r_{\text{final}} = \cos(\phi - \pi)u + \sin(\phi - \pi)W$, where $W = \vec{n} \times u$ is the tangent vector that is perpendicular to u and lies in the plane of motion (note: we subtract π in our calculation to align with the setup corresponding to our 2D plots).
- The points (x, y, z) corresponds to our r_{final} vector from which we calculate the longitude and latitude of the resulting source point, (β', ϕ') . $\beta' = \arctan 2(y, x)$, $\phi' = \arcsin(z)$.
- We now normalise (β', ϕ') to lie within $[0, 1]$. Multiplying our sky map pixels (1080 and 1920) by this we can pinpoint a pixel for each rotated ray, thus leading to a RGB value which we colour our camera pixel as.
- If $\beta = NaN$ (when our ray gets captured) we instead assign the RGB value $[0, 0, 0]$ to obtain the shadow of the black hole.

In the following visualisation we will generate 1000 rays for our data set due to limited processing power and interpolate for the rest.

A key detail to note is that for the most accurate and cinematic image, we would want to cast multiple sub-rays per pixel and average their RGB values. In reality, a single pixel, especially near the photon ring, contains significant amounts of compressed light from many different stars, and this averaging technique (anti-aliasing) physically simulates that density.

5.3 Lensing visualisation



(a) Celestial sphere.



(b) Celestial sky map.

Figure 5.3: We will demonstrate our lensing image method with this space surrounding the black hole and camera.



Figure 5.4: Resultant lensed image with $M = 20$, $r_{\text{sky}} = 1000$, $r_{\text{camera}}=800$

This star map is particularly useful as it demonstrates that even if most of the space that the camera is facing is blank, we will still see a light pattern due to lensing.

Our camera is positioned such that it would observe mostly empty space from the celestial sky, however due to gravitational lensing we have a display of the main features we would expect to see. We observe a circular shadow and a photon ring that lies close to the photon sphere. The surrounding bright ring of light comes from distant stars whose light rays follow geodesics around the black hole and deflect across large angles. Our celestial sky map presents a clustered star field surrounded by blank space, due to this we can the lensed image portrays a similar pattern "wrapped" around the black hole by its circular geometry.

5.4 Advanced properties of the Schwarzschild black hole

In our lensed image, whilst we follow a strict mathematical process applying the metric, we do not account for some very significant features we would expect to see when observing a black hole. The reason we omit some of these features is because they either do not have a major impact when considering a very distant observer, or simply because adding the feature requires more complex programming that requires significant processing power.

5.4.1 Gravitational redshift

This physical phenomenon occurs due to the significant gravitational effect of black holes and causes a frequency shift in light rays surrounding the black hole, this makes light appear "more red" towards the shadow of the black hole. The reason we expect to see a redshift for light rays whose trajectories follow closer to the event horizon is because light rays lose energy whilst escaping the gravitational well by increasing their wavelength (and thus reducing their frequency), the more energy required to escape the more of a redshift we would expect. We can see a demonstration of the redshift in figure 5.2, the asymmetry of light intensity is due to the Doppler effect caused by spin which we will discuss in 7.4.1. In our visualisation due to the distance of the camera and the type of sky map used we would find the gravitational redshift to be negligible.

5.4.2 Accretion disks

Figure 5.2 has an extra feature compared to the image we have produced, however, this feature is very significant when it comes to the appearance of a black hole. We are of course referring to the accretion disk, a ring of matter that orbits the black hole. The significance of the accretion disk is that the matter emits light through radiation, thus it accentuates the features of the black hole. We discussed the effect of the the Einstein ring, due to the close proximity of the accretion disk, we are able to see the light that emits from the top and bottom of the accretion disk (from either side of the black hole) in the poles of the black hole. This makes the Einstein ring more pronounced.

Figure 5.2 depicts a spinning Kerr black hole with a gaseous accretion disk taken directly facing the black hole. If we were to observe an accretion disk from an inclined angle we would be able to see it across the front of the black hole and feed into the lensed light around the shadow. Our main focus for this metric is to act as the foundation of our visual and mathematical understanding of black holes. As such we will not include an accretion disk and focus on the foundational ray tracing process.

Chapter 6

The Kerr Metric

Now that we have established our foundational metric, the Schwarzschild metric, and demonstrated ray tracing, we can begin to examine the Kerr metric. The Kerr metric is the metric used for the spacetime of a Kerr black hole, an uncharged, spinning, and axially symmetric black hole.

The Kerr metric is the concluding metric of this paper as this is the metric under which Kip Thorne and his team based the black hole Gargantua from the movie *Interstellar*. We will set up the mathematics for the metric and derive the properties needed to both understand what we would expect to see and how to ray trace around a Kerr black hole.

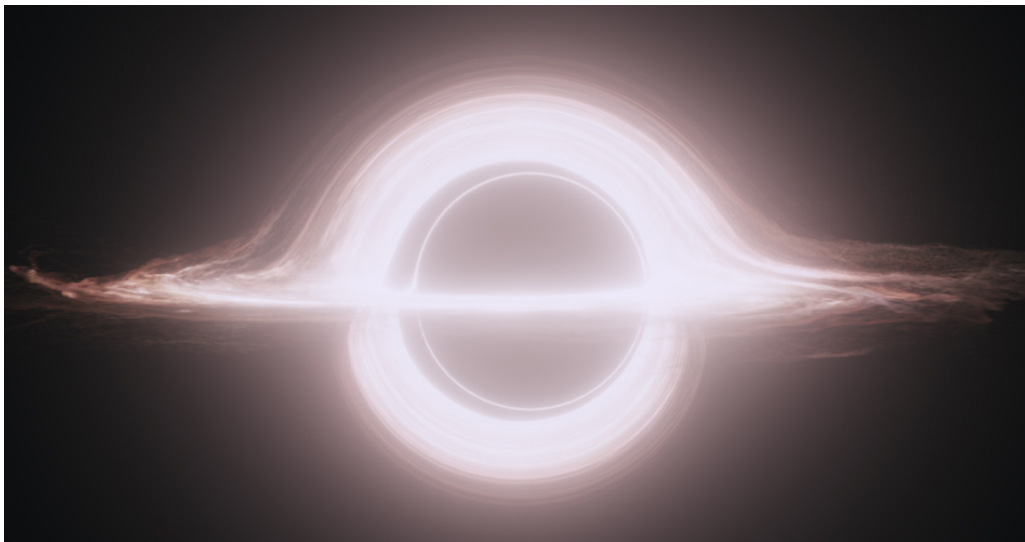


Figure 6.1: Gargantua render taken from Kip Thorne’s paper, [James et al. \[2015\]](#).

Going forward in this chapter, we will closely follow material from [[Hackmann](#), pages 78-84] for our derivations.

6.1 What is the Kerr metric?

The Kerr metric, as stated previously, is the metric for the spacetime around an uncharged, spinning, and axially symmetric black hole. It was discovered in 1963 by Roy Kerr derived from the Schwarzschild metric from chapter 4.

The metric has two coordinate system forms, Boyer-Lindquist and Kerr-Schild. We will focus purely on the Boyer-Lindquist form, as this is best used for ray tracing outside of the black hole.

The Boyer-Lindquist coordinate system, (r, θ, ϕ, t) has Cartesian geometry:

$$\begin{aligned}x &= \sqrt{r^2 + a^2} \sin \theta \cos \phi, \\y &= \sqrt{r^2 + a^2} \sin \theta \sin \phi, \\z &= r \cos \theta.\end{aligned}\tag{6.1}$$

We write the line element (spacetime interval) of the Kerr metric as:

$$\begin{aligned}ds^2 = -c^2 d\tau^2 = & - \left(1 - \frac{r_s r}{\Sigma}\right) c^2 dt^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 \\& + \left(r^2 + a^2 + \frac{r_s r a^2}{\Sigma} \sin^2 \theta\right) \sin^2 \theta d\phi^2 - \underbrace{\frac{2r_s r a \sin^2 \theta}{\Sigma} c}_{\text{coupling}} dt d\phi.\end{aligned}\tag{6.2}$$

Where:

$$\begin{aligned}r_s &= \frac{2GM}{c^2}, \\a &= \frac{J}{Mc}, \\\Sigma &= r^2 + a^2 \cos^2 \theta, \\\Delta &= r^2 - r_s r + a^2.\end{aligned}$$

Here J is the angular momentum of the black hole. In our calculations, we will choose $G = c = 1$ as we did in chapter 4 for simplicity. In this instance our metric has a signature $(-+++)$.

We can see that the main characteristics described above lie within the line element:

- It is not static: it is not invariant for time reversal, $t \rightarrow -t$, (we discussed invariance in chapter 2).
- It is stationary: it does not depend explicitly on time.
- It is axisymmetric: it does not depend explicitly on ϕ .

We express the metric tensor as:

$$\begin{aligned}g^{\mu\nu} &= \begin{bmatrix} g^{tt} & 0 & 0 & g^{t\phi} \\ 0 & \frac{\Delta}{\Sigma} & 0 & 0 \\ 0 & 0 & \frac{1}{\Sigma} & 0 \\ g^{t\phi} & 0 & 0 & g^{\phi\phi} \end{bmatrix}, \\g^{tt} &= -\frac{1}{\Delta} \left(r^2 + a^2 + \frac{2Mr a^2}{\Sigma} \sin^2 \theta \right), \\g^{t\phi} &= -\frac{2Mr}{\Sigma \Delta} a, \\g^{\phi\phi} &= \frac{\Delta - a^2 \sin^2 \theta}{\Sigma \Delta \sin^2 \theta}.\end{aligned}\tag{6.3}$$

6.2 Obtaining the equations of motion under the Kerr metric

In chapter 4 we obtained the equations of motion (EOM) for light rays under the Schwarzschild metric through the geodesic equations. For the Kerr metric we will take a more general approach to the EOM and make clear links to the previous method. Through this alternative approach we will draw links between relativistic properties of particles and the mathematical physics we have developed thus far.

Our goal is to develop a set of ODEs for the EOM under this metric, as before, we want to simplify these ODEs as much as possible and decouple them. We begin by looking at the generalised momenta to obtain two constants of motion (COM):

$$p_t = g_{tt} \frac{dt}{d\tau} + g_{t\phi} \frac{d\phi}{d\tau} = E, \quad (\text{conserved energy}) \quad (6.4)$$

$$-p_\phi = -g_{\phi\phi} \frac{d\phi}{d\tau} - g_{t\phi} \frac{dt}{d\tau} = L_z. \quad (\text{axial angular momentum}) \quad (6.5)$$

We obtained the equivalent constants for the Schwarzschild metric in Chapter 4 through the geodesic equations, which required significantly longer derivation. The question posed by this new, faster approach is why not take the same approach for the Schwarzschild metric?

The answer is that we could have derived the constants of motion for the Schwarzschild EOM with this method as ultimately, the geodesic equations adhere to these conserved momenta. Through originally deriving these constants via the geodesic function, we ultimately grasp the concept of how the curvature affects the light ray in order to ensure these constants remain constant. Now, going forward, we can see how robust general relativity is by uniting the conceptual idea with its computational counterpart.

We can directly establish the link between generalised momenta and the geodesic equation (note: this is not necessary knowledge for our derivations, but helps to solidify our understanding of utilising conserved momenta) if we consider an equivalent Euler-Lagrange approach to what we followed in section 3.1.

$$\begin{aligned} p_\mu &= \frac{\partial l(\mathbf{x}, \dot{\mathbf{x}}, \tau)}{\partial \dot{x}^\mu}, \text{ we use the Lagrangian } l = \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu, \text{ which gives the hamiltonian formalism:} \\ p_\alpha &= \frac{\partial}{\partial \dot{x}^\alpha} \left(\frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \right) = \frac{1}{2} g_{\mu\nu} \left(\frac{\partial \dot{x}^\mu}{\partial \dot{x}^\alpha} \dot{x}^\nu + \frac{\partial \dot{x}^\nu}{\partial \dot{x}^\alpha} \dot{x}^\mu \right) = \frac{1}{2} g_{\mu\nu} (\delta_\alpha^\mu \dot{x}^\nu + \delta_\alpha^\nu \dot{x}^\mu) \\ \implies p_\alpha &= \frac{1}{2} (g_{\alpha\nu} \dot{x}^\nu + g_{\mu\alpha} \dot{x}^\mu) = \frac{1}{2} (2g_{\alpha\nu} \dot{x}^\nu) = g_{\alpha\nu} \dot{x}^\nu. \end{aligned}$$

This links to the geodesic equations as the Euler Lagrange equation gives

$$\frac{\partial}{\partial \lambda} p_\alpha = \frac{\partial l}{\partial x^\alpha} \implies \frac{\partial}{\partial \lambda} (g_{\alpha\nu} \dot{x}^\nu) = \frac{1}{2} \frac{\partial g_{\mu\nu}}{\partial x^\alpha} \dot{x}^\mu \dot{x}^\nu.$$

If we expand the left hand side using the chain rule we get:

$$\frac{\partial}{\partial \lambda} g_{\alpha\nu} = \frac{\partial g_{\alpha\nu}}{\partial x^\beta} \dot{x}^\beta \implies \dots \implies \ddot{x}^\mu + \Gamma_{\rho\sigma}^\mu \dot{x}^\rho \dot{x}^\sigma.$$

We have shown how we can link our general relativistic approach to obtaining COM with the geodesic approach, but we still need more COM in order to simplify and decouple the EOM under the Kerr metric.

So far we have two COM for a four-dimensional system, we still require another two COM in order to simplify the EOM into ODEs. As such, we begin by deriving the **Carter constant**, a constant of importance to analytically computational EOM.

We follow Brandon Carter's approach utilising the Hamilton-Jacobi equation to obtain conserved quantities:

$$2\frac{\partial S}{\partial \tau} = g^{\mu\nu} \frac{\partial S}{\partial x^\mu} \frac{\partial S}{\partial x^\nu}.$$

Brandon Carter decided to use this specific equation as it reveals hidden symmetries, specifically isolating killing tensors (symmetries that yield COM along geodesics) (general tensor theory can be found in [Schutz](#)[section 3]). He proposed an ansatz that allows for the separability of the action, S , into a summation of its dependent variables (known as additive separability).

$$S = \frac{1}{2}\epsilon\tau - Et + L_z\phi + S_r(r) + S_\theta(\theta).$$

Where $\epsilon = g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ is a constant dependent on the particle's size. **Note:** $\epsilon = 0$ for null geodesics, so we do not require its inclusion going forward, however, we will still include it for the purpose of recognition.

By proposing this action, Carter was able to apply the Hamilton-Jacobi equation to develop the following equation:

$$\epsilon a^2 \cos^2 \theta + \left(\frac{\partial S}{\partial \theta}\right)^2 + \frac{1}{\sin^2 \theta} (aE \sin^2 \theta - L_z)^2 \quad (6.6)$$

$$= -\epsilon r^2 - (r^2 + a^2 - 2Mr) \left(\frac{\partial S}{\partial r}\right)^2 + \frac{1}{(r^2 + a^2 - 2Mr)} ((r^2 + a^2)E - aL_z)^2. \quad (6.7)$$

Here we have an equation with the left hand side being entirely dependent on θ and r on the right hand side. We take both these sides to be the COM we are missing, the Carter constant. Thus (6.6) and (6.7) = K , K being the Carter constant.

Note: in a lot of modern mathematics involving the Kerr metric, we often see the use of the Carter constant, $Q = K - (aE - L_z)^2$. The reason we have an adapted Carter constant, Q , is because it makes the mathematics of geometrical interpretations much simpler. It is important to make aware of this, as later we will use it in some derivations.

Now that we have the necessary constants of motion, we can start developing our equations of motion. Before we expressed the hamiltonian formalism for canonical momenta,

$$\frac{\partial l}{\partial \dot{x}^\mu} = p_\mu = \frac{\partial S}{\partial \dot{x}^\mu}.$$

As such, we can express our generalised momenta

$$p_r = g_{rr}\dot{r} = \frac{\Sigma}{\Delta}\dot{r}, \quad \text{as } \frac{\partial S}{\partial r} = \frac{\Sigma}{\Delta} \frac{dr}{d\tau}.$$

6.2 Obtaining the equations of motion under the Kerr metric

Now we use the right hand side of the Carter constant, (6.7), which we can express as:

$$\begin{aligned}
& - \left(\frac{dS_r}{dr} \right)^2 \Delta + (\text{other } r \text{ terms}) = K \\
& \implies - \left(\frac{dS_r}{dr} \right)^2 \Delta = K + \epsilon r^2 - \frac{1}{\Delta} \left((r^2 + a^2)E - aL_z \right)^2 \\
& \implies \left(\frac{dS_r}{dr} \right)^2 = \frac{1}{\Delta^2} \left((r^2 + a^2)E - aL_z \right)^2 - \frac{1}{\Delta} (\epsilon r^2 + K),
\end{aligned}$$

Thus,

$$\Sigma^2 \left(\frac{dr}{d\tau} \right)^2 = \left((r^2 + a^2)E - aL_z \right)^2 - \Delta(\epsilon r^2 + K) := R(r)$$

is our first EOM.

The same process can be applied to the left hand side of the Carter constant, (6.6), to get our second EOM:

$$\Sigma^2 \left(\frac{d\theta}{d\tau} \right)^2 = \left(K - \epsilon a^2 \cos^2 \theta \right) - \frac{1}{\sin^2 \theta} \left(aE \sin^2 \theta - L_z \right)^2 := \Theta(\theta).$$

In order to obtain $\frac{d\phi}{d\tau}$ and $\frac{dt}{d\tau}$, we do not require the ansatz, rather we go back to momenta:

$$\begin{aligned}
g_{tt}\dot{t} + g_{t\phi}\dot{\phi} &= E \\
g_{\phi t}\dot{t} + g_{\phi\phi}\dot{\phi} &= -L_z
\end{aligned}$$

$$\implies \begin{pmatrix} g_{tt} & g_{t\phi} \\ g_{\phi t} & g_{\phi\phi} \end{pmatrix} \begin{pmatrix} \dot{t} \\ \dot{\phi} \end{pmatrix} = \begin{pmatrix} E \\ -L_z \end{pmatrix}.$$

We use the inverse metric, $g^{\mu\nu}$, to get:

$$\begin{aligned}
\Sigma \frac{d\phi}{d\tau} &= \frac{a}{\Delta} \left((r^2 + a^2)E - aL_z \right) - \frac{1}{\sin^2 \theta} \left(aE \sin^2 \theta - L_z \right), \\
\Sigma \frac{dt}{d\tau} &= \frac{r^2 + a^2}{\Delta} \left((r^2 + a^2)E - aL_z \right) - a \left(aE \sin^2 \theta - L_z \right).
\end{aligned}$$

Now we have equations of motion under the Kerr metric; however, we cannot claim that these equations are uncoupled as $\Sigma = r^2 + a^2 \cos^2 \theta$ is a function of both r and θ . To eliminate Σ from our equations and thus decouple our EOM, we introduce **Mino time**, λ . The concept of Mino time was introduced by Yasushi Mino in his paper, [Mino \[2003\]](#)

The relation is

$$\frac{d\tau}{d\lambda} = \Sigma.$$

Mino time, λ , (not to be confused with the affine parameter we have previously used for our geodesic equations) tells us how the proper time of a particle is warped by the black hole's rotation.

Ultimately, the equations of motion for a light ray under the Kerr metric (setting $\epsilon = 0$) are:

$$\left(\frac{dr}{d\lambda}\right)^2 = \left((r^2 + a^2)E - aL_z\right)^2 - \Delta K := R_K(r), \quad (6.8)$$

$$\left(\frac{d\theta}{d\lambda}\right)^2 = K - \frac{1}{\sin^2 \theta} \left(aE \sin^2 \theta - L_z\right)^2 := \Theta_K(\theta), \quad (6.9)$$

$$\frac{d\phi}{d\lambda} = \frac{a}{\Delta} \left((r^2 + a^2)E - aL_z\right) - \frac{1}{\sin^2 \theta} \left(aE \sin^2 \theta - L_z\right), \quad (6.10)$$

$$\frac{dt}{d\lambda} = \frac{r^2 + a^2}{\Delta} \left((r^2 + a^2)E - aL_z\right) - a(aE \sin^2 \theta - L_z). \quad (6.11)$$

6.3 Features of the Kerr black hole

Before we can begin ray tracing under the Kerr metric we need to establish important properties of the Kerr black hole. As in chapter 4, we require the radial position of the event horizon in order to apply conditional events to our backwards integration.

6.3.1 The event horizon

It is vital that we find the radius of the event horizon in order to omit any light rays that get captured by the black hole as we did with Schwarzschild black hole.

We derive the Kerr black hole's event horizon by considering the divergence of the radial component under this metric; by doing this, we are isolating the radius where the coordinate system fails due to the geometry, [Wikipedia](#) [f].

$$\begin{aligned} g_{rr} \rightarrow \infty &\implies \frac{1}{g_{rr}} \rightarrow 0, \\ \text{thus } g^{rr} \rightarrow 0 &\implies \frac{\Delta}{\Sigma} = 0. \end{aligned} \quad (6.3)$$

So we find

$$\begin{aligned} \Delta = r^2 - 2Mr + a^2 = 0 &\implies r = \frac{2M \pm \sqrt{4M^2 - 4a^2}}{2} \\ &\implies r = M \pm \sqrt{M^2 - a^2} \end{aligned} \quad (6.12)$$

Here we can see there are actually two possible event horizons for the Kerr black hole. We call the additive root, r_+ , the **outer horizon** which relates to the primary event horizon. Whereas the inner horizon, r_- , relates to the Cauchy horizon, which is a highly unstable boundary deep inside the black hole. As such we will only require the outer horizon for our ray tracing process. We can see that if the Kerr black hole has no spin ($a = 0$), then $r_+ = 2M := r_s$. This imposes the Schwarzschild radius, as we would expect.

6.3.2 The photon sphere

The photon sphere, as with the Schwarzschild metric, can be derived from the radial ODE, $R_K(r)$ (6.8).

The nature of the photon sphere dictates that a photon in this orbit will remain at a constant radius and be unable to escape thus we can deduce,
 $R_K = \dot{R}_K = 0$.

We previously mentioned that we can use a more geometrical interpretation for the Carter constant, $Q = K - (aE - L_z)^2$. We will be utilising this for our derivation of the photon sphere, in doing so we will see that the mathematics become simpler when we consider a geodesic that lies entirely in the equatorial plane, $\theta = \frac{\pi}{2}$.

$$R_K(r) = \left((r^2 + a^2)E - aL_z \right)^2 - \underbrace{\Delta \left[Q + (aE - L_z)^2 \right]}_{=K} = 0,$$

$$R_K'(r) = 4rE \left((r^2 + a^2)E - aL_z \right) - \Delta' \left[Q + (aE - L_z)^2 \right] = 0. \quad (Q \text{ is a COM so its derivative is } 0)$$

We will simplify these equations by rescaling with respect to energy to reduce dimensions.

$$\gamma = \frac{L_z}{E}, \quad \eta = \frac{Q}{E^2}.$$

Substituting in these parameters and solving simultaneously we get:

$$\begin{aligned} (r^2 + a^2) - a\gamma &= \frac{4r\Delta}{2r - 2M} \\ \implies 2r^3 - 6Mr^2 + r(2a^2 + 2a\gamma) + 2Ma^2 - 2Ma\gamma &= 0 \\ \implies \gamma &= \frac{r^2(r - 3M) + a^2(r + M)}{a(M - r)}. \end{aligned} \tag{6.13}$$

Now we have an expression for the scaled angular momentum required for a photon to be held in orbit.

We can substitute this back into the equation for $\dot{R}_K(r)$ and rearrange to get an expression for η :

$$\begin{aligned} \eta &= \frac{4r^2\Delta}{2(r - M)^2} - \left(\frac{r(r^2 - 3Mr + 2a^2)}{a(M - r)} \right)^2 \\ &= \frac{r^3 [4Ma^2 - r(r - 3M)^2]}{a^2(r - M)^2}. \end{aligned} \tag{6.14}$$

We introduce a theorem taken from [Hackmann](#)[pages 81-82].

Theorem 6.1. *If a geodesic lies entirely in the equatorial plane $\theta = \frac{\pi}{2}$ then the modified Carter constant $Q = K - (aE - L_z)^2$, under the Kerr metric, is zero.*

The proof follows from the ODE (6.9), we can simply observe that given $\theta = \frac{\pi}{2}$, $\left(\frac{d\theta}{d\lambda}\right)^2 = 0 = K - (aE - L_z)^2 := Q$.

Applying this theorem allows us to find the radii at which the photon sphere occurs:

$$\begin{aligned} \implies r^3 [4Ma^2 - r(r - 3M)^2] &= 0 \\ \implies r^3 - 6Mr^2 + 9M^2r - 4Ma^2 &= 0. \end{aligned}$$

$$\begin{aligned} r_1 &= 2M \left[1 + \cos \left(\frac{1}{3} \arccos \left(\frac{a}{M} \right) \right) \right], \\ r_2 &= 2M \left[1 + \cos \left(\frac{1}{3} \arccos \left(-\frac{a}{M} \right) \right) \right]. \end{aligned}$$

The two radii r_1 and r_2 , respectively represent the retrograde photon sphere and prograde photon sphere radial positions. These two photon spheres are what determine the "D shape" asymmetric shadow of a Kerr black hole. The retrograde photon sphere embodies the photons that orbit against the black hole's spin and thus lose their energy faster and are captured more easily, this creates a greater shadow. The prograde photon sphere embodies the light rays that follow the same direction as the black hole's spin and thus "surf" the spacetime, they can get extremely close to the event horizon before finally falling in making the shadow appear smaller.

Chapter 7

Ray Tracing under the Kerr metric

Now we have derived the two major features of the Kerr black hole and in doing so demonstrated the usefulness of the geometrical version of the Carter constant, Q . We can now move on to develop some ray tracing visuals for the Kerr metric as we did for the Schwarzschild black hole in chapter 5.

7.1 Modelling 2D light ray trajectories

We can now follow the same process we used in chapter 5 to generate a trajectory plot along with a plot of respective angles. We consider the camera to lie on the equatorial plane such that $\theta = \frac{\pi}{2}$.

Note: our geodesic equations are special in this case due to the camera being on the equatorial plane. This condition allows us to limit the geodesics to a single plane for our 2D plot.

We in turn only focus on the equations $R_K(r)$, $\dot{R}_K(r)$ and $\frac{d\phi}{d\lambda}$ for our Solve_ivp function (6.8).

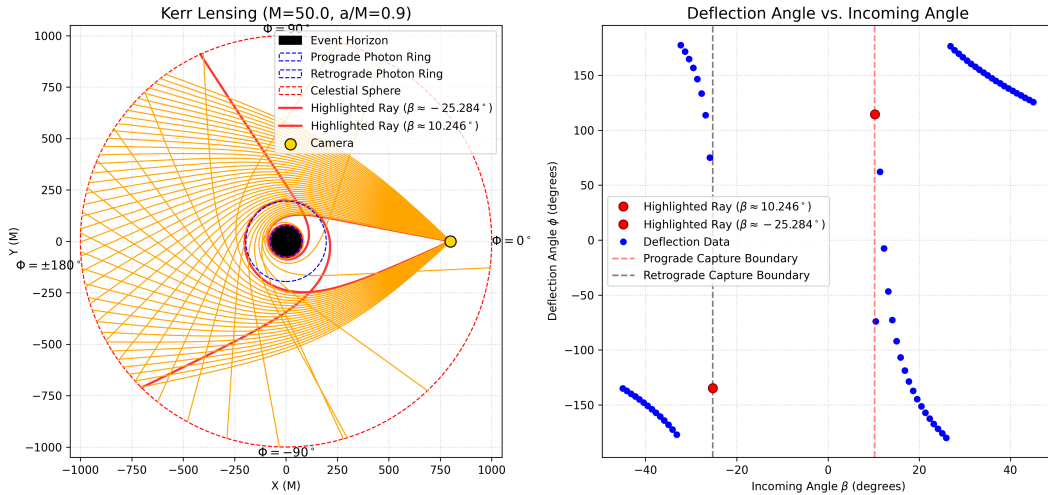


Figure 7.1: Plots with $r_{\text{camera}} = 800$, $r_{\text{sky}} = 1000$ and 100 rays.

We have once again limited our incoming ray angles to the camera to $\beta \in [-45^\circ, 45^\circ]$ and the deflection angle (or source angle) to $\phi \in [-180^\circ, 180^\circ]$ (**Note:** we interpret this 2D plot in polar coordinates not Boyer-Lindquist).

Here we have set the spin ratio of the black hole, $\frac{a}{M}$, to be positive which produces a clockwise

spin about the vertical axis, relative to the camera.

We are able to visually demonstrate the two photon spheres of the Kerr black hole. Here we have used a greater mass to signify the unique properties of the Kerr metric; however, if we were to use the same mass, $M = 20$, as we did with the Schwarzschild demonstration, we would see a greater gap between the data, signifying a bigger shadow due to the spin property as we mentioned at the end of the previous chapter. We have two highlighted rays that lie close to our respective photon spheres. The 2D plot clearly demonstrates the asymmetry of the shadow and thus the significance of the spin. We can see a semblance of the Schwarzschild metric as the photons reach more significant angles, and thus are impacted less by the spin and begin to exhibit a behaviour more fitting of the Schwarzschild metric.

7.2 What we would expect to see

We know from our mathematical derivations that the Kerr metric follows closely from the Schwarzschild metric; however, the equations of motion, events horizons and photons spheres are far more complex. The question is how does this differentiate from the Schwarzschild black hole visual?

We derived two photon spheres and briefly discussed how this would lead to an asymmetric shadow. The shadow of Kerr black hole is dependent on its spin to mass ratio, $\frac{a}{M}$. A high ratio will result in a more asymmetric shadow, typically a "D shadow" due to the fact that one side, corresponding to the light rays going against the spin, will have a greater shadow region.

The opposite side of the shadow appears almost truncated in comparison to the symmetric shadow of a Schwarzschild black hole due to the fact that light rays "surfing" the spin lose less energy and thus are less likely to fall into the black hole, resulting in a smaller shadow. As the spin is based around one axis we only see asymmetry in that axis, that being the vertical axis for the observer.

This unique shadow is clearly demonstrated by figure 7.1 where both the trajectory plot and scatter plot demonstrate an asymmetric shadow. The scatter plot clearly shows the incoming angle difference for escaped rays on either side of the black hole with a clockwise spin, as well as the different angle boundaries for the unique photon spheres.

Due to the spin of the Kerr black hole, the lensing effect is much more aggressive. We discussed the lensing effect that forms the Einstein ring in section 5.2, however with the Kerr black hole we see the Einstein ring surrounding the black hole to also be asymmetric along with the shadow as we would expect.

The spin of the black hole also leads to a significant phenomenon which is not visually apparent in our plot or the lensed image we will generate, however, it is an important property for understanding the Kerr black hole, especially when we refer to the advanced features later in this chapter (7.4). This effect is called **frame dragging** and affects particles (we introduced the concept of orbiting matter in 5.4.2) within the ergosphere.

The **ergosphere** is a region that lies beyond the event horizon where it is characterised as the sphere where "the rotational velocity of the surrounding space is dragged along with the velocity of light" -[Wikipedia](#) [f]. This region has major significance because it will cause particles to gain energy to the point where they emit radiation and may even escape the event horizon they

otherwise would have fallen into. Another effect on lensing due to spin is the potential creation or annihilation of image pairs, this concept follows caustics which is a concept further discussed in [James et al. \[2015\]](#), but this level of depth is not necessary for understanding our images due to the stationary sky map and camera.

7.3 Lensing visualisation

In order to see the significance of the Kerr metric we will draw comparisons to the Schwarzschild black hole as we have done above with our 2D plot. However, in order to create a lensed image of the same celestial sphere we used in chapter 5,

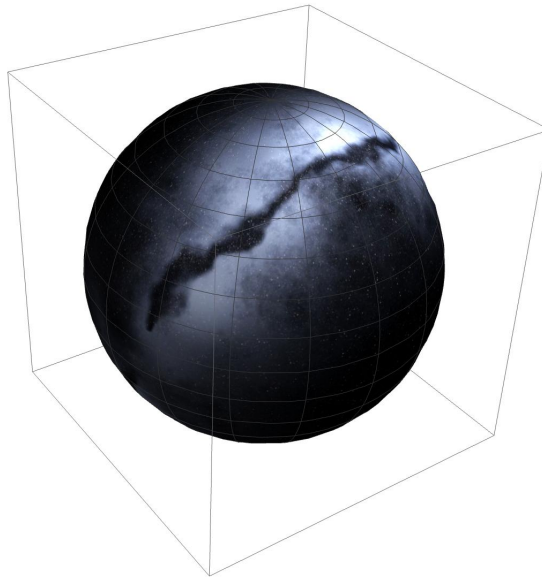


Figure 7.2: Celestial sphere we wish to use as with the Schwarzschild metric.

we have to adapt our method. We can no longer use the property of symmetry in order to represent a full 2D grid of pixels. Considering the geodesics not being limited to a single plane, we need to adapt our ray trajectory data generation to account for the vertical angle of our varying rays, so we bring back our θ component and include our $\Theta_K(\theta)$ (6.9) equation in our program.

When we generated a lensed image for the Schwarzschild metric, we switched between polar and Cartesian coordinates to map our pixels from the camera to their corresponding pixels on the sky map. For the Kerr metric, we adopt **Bardeen** coordinates (α, β) to "bridge" the gap between the camera lens and the celestial sphere. These coordinates represent **impact parameters** (displacements of a light ray from the black hole's centre as seen by a distant observer). Because our camera is placed at a near-infinite distance, these displacements are directly proportional to the angles hitting the camera (utilising small angle approximations) for rays whose path follows close to the black hole's horizon. This allows us to integrate rays with complex geodesics more efficiently by avoiding trigonometric calculations.

This method simplifies the initialisation of our rays: the horizontal displacement becomes the

axial angular momentum, $L_z = -\alpha$; while the vertical displacement, β , defines the scaled Carter constant, $\eta = \frac{Q}{E^2}$, to be $\eta = \beta^2$ and $K = \beta^2 + (L_z - a)^2$. This approach is very efficient for our angles close to 0 degrees (rays heading directly towards the black hole), however, if we want a full 90 degree field of view we have to also consider larger angles, which we cannot apply the small angle approximation to.

For such angles we instead use the coordinates ($\alpha = r_{cam} \sin(\theta_{obs}) \cos(\phi_{obs})$, $\beta = r_{cam} \sin(\phi_{obs})$), where θ_{obs} (this is the same angle as β from our 2D plot, we change notation here to match Bardeen notation) is the incoming rays horizontal angle and ϕ_{obs} is the vertical angle between the ray and x-y plane (as opposed to the angle between the ray and spin axis in Boyer-Lindquist geometry).

Bardeen's method is also used in the DNGR code, however, as DNGR is used for a close image of the black hole their method involves more complex techniques which we will see along with the Boyer-Lindquist coordinate system.

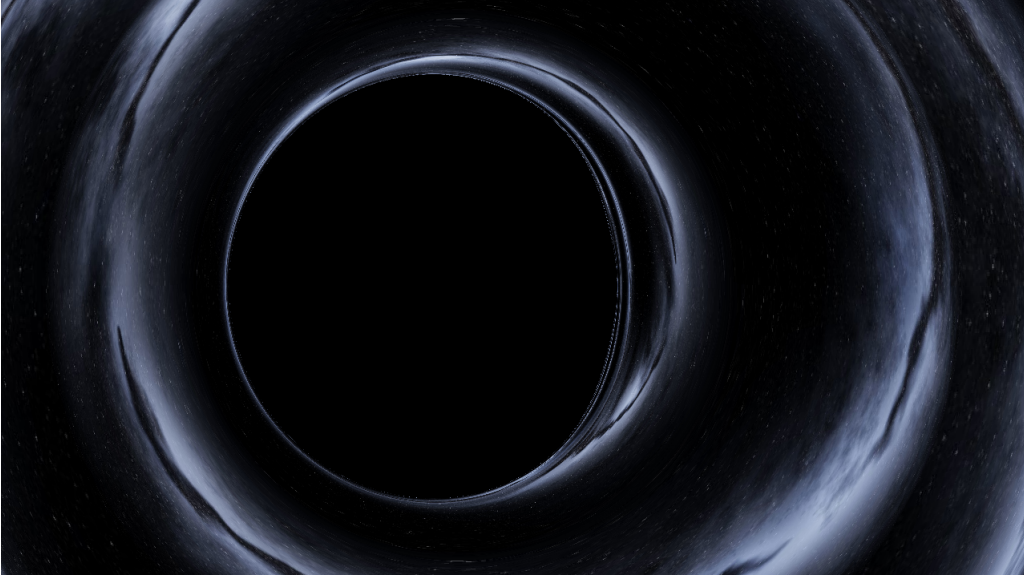


Figure 7.3: A 720p lensed image using: $r_{camera} = 900$, $r_{sky} = 1000$, $M = 50$, $\frac{a}{M} = 0.99$.

Here we use different parameter values to the Schwarzschild visualisation (and increased the spin ratio used in our 2D Kerr plot) to exemplify the features of the Kerr black hole. We can see the expected asymmetric shadow along with the asymmetric lensed star field.

An important aspect of our visualisation to note is that due to the asymmetry of the Kerr black hole, interpolation is less reliable for imaging and so we require the integration of significantly many more rays than we did with the Schwarzschild metric. Due to the density of rays needed for a reasonable quality image we require a lot of processing power and time to render an image. A way to combat the intensity of the render is to use C++ as a coding language that allows for GPU focused processing and to adopt ray bundling integration which we will discuss later. However, we have chosen to focus on the mathematical aspect of black holes and continue to use python for a simplistic lensing image without the additional features we will discuss and see when we look at the black hole Gargantua.

7.4 Advanced properties of the Kerr black hole

We can observe a colour map to portray the properties we have discussed.

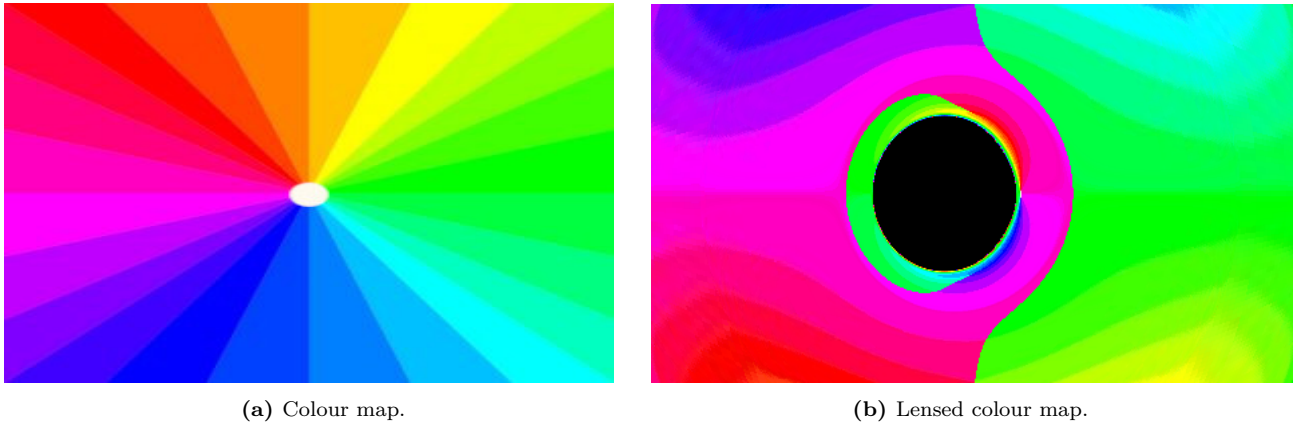


Figure 7.4: Using: $r_{\text{camera}} = 900$, $r_{\text{sky}} = 1000$, $M = 50$, $\frac{a}{M} = 0.99$ in accordance with our previous lensed image.

7.4 Advanced properties of the Kerr black hole

Just like the visualisation of the Schwarzschild black hole, our visualisation for the Kerr black hole has omitted key features either due to insignificance in results or due to lack of computer processing power.

7.4.1 Gravitational redshift, the Doppler effect and Doppler dimming

A Kerr black hole expectedly displays gravitational redshift which we discussed in section 5.4.1. For the same reason we have, once again, chosen to omit redshift in our visualisation. However there is an additional phenomenon that we would expect to see when observing a Kerr black hole, the Doppler effect.

The Doppler effect is the apparent change in the frequency of a wave relative to the motion of the source or observer. In short, the movement of a light ray source influenced by the spin of the black hole will cause a change in frequency and thus colour of the light emitted.

Doppler dimming (also known as Relativistic beaming) is a less known phenomenon as it usually only occurs with celestial bodies. This effect occurs when a light source moves at relativistic speeds (close to the speed of light), like matter in the ergosphere of a Kerr black hole, causing the emitted light to "focus" in the direction of motion. This results in the asymmetric light intensity depending on the observer's position.

7.4.2 The accretion disk

An accretion disk is more substantial for a Kerr black hole than a Schwarzschild black hole. Figure 8.1, the Gargantua image taken from Interstellar, has the distinct ring of matter that lenses around the black hole just as we established in Chapter 5. We described the ergosphere for a Kerr black hole, visually the ergosphere is portrayed by the accretion disk.

The matter orbiting the black hole is accelerated to the speed of light causing a significant gain in energy, as such we can observe radiation emission. The frequency range of the radiation depends on the mass of the black hole and the temperature of the matter "Gas heated to

20 million degrees Fahrenheit as it spirals toward the black hole glows in low-energy, or soft, X-rays." - [Nasa](#). We previously defined the Doppler effect and Doppler dimming, it should now be very clear how the accretion disk acts as a vast source of light demonstrating these effects. At extreme temperatures with the Doppler effect and redshift of black body radiation we see a wide spectrum of radiation emitted from the accretion disk. We have seen the Doppler effect and redshift in figure 5.2. Additionally, the intensity of the light emitted by the accretion disk is dependent on the spin of the black hole and the observer's position. Visually we expect to have an extremely bright accretion disk that demonstrates asymmetry in both the colour and intensity of light.

Chapter 8

Interstellar's Gargantua



Figure 8.1: Black hole Gargantua used in the movie Interstellar.

The movie interstellar has its own Kerr black hole visualisation, the black hole called Gargantua. This black hole is a stunning movie spectacle that influenced the perception of a black hole for millions of people across the world.

The Kerr black hole generated for the movie follows the mathematics we established in Chapter 6. The team at Double Negative Visual Effects, collaborating with Kip Thorne, created their own code model called the double negative gravitational renderer (DNGR). In this chapter we will follow from the article published in 2015, [James et al. \[2015\]](#) following the movie's release in 2014. We will explore the DNGR coding process from a mathematical perspective and use the knowledge we have accumulated so far to criticise the visual generated for Interstellar.

8.1 DNGR code process

In the previous chapters we have developed clear understanding of what ray tracing is and allows us to do. We established fundamental concepts in general relativity and special relativity required to comprehend what gravitational lensing is and how it occurs. We explored the concept of ray tracing in enough detail to generate simple lensing images of both the Schwarzschild and Kerr metrics from the perspective of a distant observer/camera.

How exactly does our method of generating visual images of black holes compare to DNGR?

DNGR goes beyond our method by creating a visualisation of a close by observer rather than producing a single distant image as we have done. At a close by range, the camera will be influenced by the black hole meaning it will experience relativistic aberration (the same principle as Doppler dimming from section 7.4.1 in which light is focused by motion, also known as beaming). This introduces much more mathematical complexity, which, in order to tackle this DNGR introduces a locally non-rotating observer, also called the fiducial observer or FIDO, to act as a frame of reference for the moving camera and surrounding space.

Computationally, DNGR handles bundles of light rays rather than individual light rays and applies averaging RGB values as we discussed in section 5.3. We previously discussed how using C++ allows for more GPU intensive processing; however, whilst the DNGR code was built in C++, it actually prioritised ray bundling which required a CPU based render farm for smooth integration.

It will come as no surprise to say that the DNGR coding model goes above and beyond what we have covered in this paper, however, that is not due to lack of understanding or knowledge. The DNGR code required immense processing power, as expected, to generate IMAX quality renders of a black hole. Unfortunately we are unable to generate such stunning visuals on a personal computer.

In our Kerr visualisation we took the approach of using Bardeen coordinates (section 7.3) to calculate our integrals, this was effective but made simple by considering the camera to lie on the equatorial plane. How does DNGR, which we mentioned uses James Bardeen's method as well, approach the challenge of a non-stationary, close by perspective not limited to the equatorial plane of a Kerr black hole?

We now refer back to the FIDO (a locally non-rotating observer). We follow [James et al. \[2015\]](#)[Appendix A.1 and pages 6-7]. The FIDO allows us to first consider the motion of the camera, which experiences frame dragging, and transform the vector of the camera's local sky (which we consider the screen when isolating the field of view). Transforming the vector relative to the FIDO allows us to eliminate any error caused by the frame dragging (and in turn relativistic aberration) and the close distance of the camera before we integrate the light ray trajectories. Now the ray calculations can be made in the FIDO's frame of reference in the same way, using Bardeen's equations but with a different set of equations of motion along with the translated initial conditions. The equations of motion used in DNGR are described as a "super-Hamiltonian version of the ray equations" in the paper which allow for better ray integrations considering the accretion disk. The DNGR code then accounts for any blue shift of the light.

8.1.1 Mathematical approach of DNGR

We have explained in a simple but lengthy manner how DNGR approaches its image generation, now we will show the mathematics responsible for it following from the paper's appendix. We will reduce the method's detail to our current level of understanding.

We begin in Boyer-Lindquist coordinates (t, r, θ, ϕ) (6.1). We have the FIDO "whose 4-velocity is orthogonal to the surfaces of constant t ", which means the position of the FIDO is fixed and

adopt their notation upon reading.

$$\begin{aligned}
\varpi &= \frac{\sqrt{(r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta} \sin \theta}{\sqrt{\Sigma}}, & \omega &= \frac{2ar}{(r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta}, \\
\alpha &= \frac{\sqrt{\Sigma \Delta}}{\sqrt{(r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta}}, & E_F &= \frac{1}{\alpha + \omega \varpi n_{F\hat{\phi}}}. \\
p_r &= E_F \frac{\sqrt{\Sigma}}{\sqrt{\Delta}} n_{F\hat{r}}, & p_\theta &= E_F \sqrt{\Sigma} n_{F\hat{\theta}}, \\
p_\phi &= E_F \varpi n_{F\hat{\phi}}.
\end{aligned}$$

These equations all look very complex but E_F is the energy of the ray measured by the FIDO, this just means that we have found the transformed momenta components.

We then use these to find our scaled Carter constant, η , and axial angular momentum, L_z .

$$L_z = p_\phi, \quad \eta = p_\theta^2 + \cos^2 \theta \left(\frac{L_z^2}{\sin^2 \theta} - a^2 \right).$$

Here you may be able to spot the similarity to the Bardeen equations we stated in section 7.3. This is because as $r \rightarrow \infty$:

$$\begin{aligned}
\alpha &\rightarrow 1, & \omega &\rightarrow 0, & \varpi &\rightarrow r \sin \theta \\
\Delta &\rightarrow r^2.
\end{aligned}$$

The energy E_F also matches our energy at infinity which we have standardised to 1. We also find that $p_\phi \rightarrow r \sin \theta n_{F\hat{\phi}}$ which at infinity is equivalent to the horizontal displacement from the spin axis which means $L_z = \alpha_{\text{Bardeen}}$ for a camera near infinity. Then we are left with the expression for the scaled Carter constant, $\eta = p_\theta^2 + \cos^2 \theta \left(\frac{L_z^2}{\sin^2 \theta} - a^2 \right) \rightarrow p_\theta^2$ as we previously assumed to be on the equatorial plane. $p_\theta \rightarrow r n_{F\hat{\theta}}$ is the vertical displacement, β_{Bardeen} , at infinity. So we are left with $\eta = \beta_{\text{Bardeen}}^2$ which also agrees with what we established for our own visualisation method.

The pair $\{L_z, \eta\}$ is then used to determine whether the ray hits the event horizon or celestial sphere. We compare the values to the conditional η (6.14) and L_z (6.13) we derived for the photon sphere determined at the radius of the event horizon at given by (6.12).

The process ends with calculating the ray trajectories through backwards integration for any rays that hit the celestial sphere. We will not delve into their "super-Hamiltonian version for the ray equations" as the method is clear without them. They are ultimately left with a resulting set of points (θ', ϕ') being the source points for the ray. They perform this process for ray bundles which they detail further in the paper and compute the light's blueshift through the equation:

$$\frac{f_c}{f'} = \left(\frac{\sqrt{1 - \beta^2}}{1 - \beta N_y} \right) \left(\frac{1 - L_z \omega}{\alpha} \right).$$

This fraction of frequency perceived by the camera over frequency at the source is used to adjust the light ray's frequency for the final visual.

8.2 What can we see in Gargantua?

Let us refer back to figure 8.1. This visualisation in *Interstellar* displays the main features of the Kerr black hole: An asymmetric shadow, an Einstein ring, a photon sphere, an accretion disk and a variation in light intensity. At a close distance we are also able to observe frame-dragging in the movie. However, this Kerr black hole is not as realistic as we would expect.

The reality of the black hole Gargantua is that some major properties were altered to make it more visually appealing; because, in the end, the main purpose of the black hole was to serve in the plot of a movie.

So what is different about this black hole?

The team behind this visual decided that for a more appealing black hole image it would be better to reduce the spin of the black hole and selectively alter the Doppler effect. In Thorne's paper it is stated "for visual purposes Nolan and Franklin slowed the spin to $a/M = 0.6$ ", this explains why the shadow of the black hole is not as asymmetric as that of our own visualisation. For the Doppler effect, the frequency ratio we discussed previously accounts for both redshift and the Doppler effect, however, for the final render of Gargantua the film producers decided to omit Doppler induced colour shifting. If they had decided to keep the full Doppler effect we would expect to see a black hole with asymmetric blue shifting. They also omitted the Doppler dimming for the same reason, otherwise we would expect the accretion disk to have asymmetric intensity.

It is also worth mentioning that in their paper they stated the use of "a disk created for *Interstellar* by Double Negative artists rather than created by solving astrophysical equations" for their accretion disk. The accretion disk itself emits red light in the movie, this is due to the temperature of the disk. In reality the temperature of an accretion disk would be extremely hot and would release high intensity radiation. We would expect the accretion disk to emit more blue and white light at normal temperatures but due to the movie requiring the situation where a spaceship is able to enter the black hole, the producers needed the temperature to be significantly decreased. The decrease in temperature resulted in a red spectrum of light.

Ultimately, the variation in light intensity that we for Gargantua (particularly more intense closer to the shadow) is primarily generated by their own beaming effect for visual appeal. They also included their own simulated lens flare for the accretion disk, [James et al. \[2015\]](#)[section 4.2].

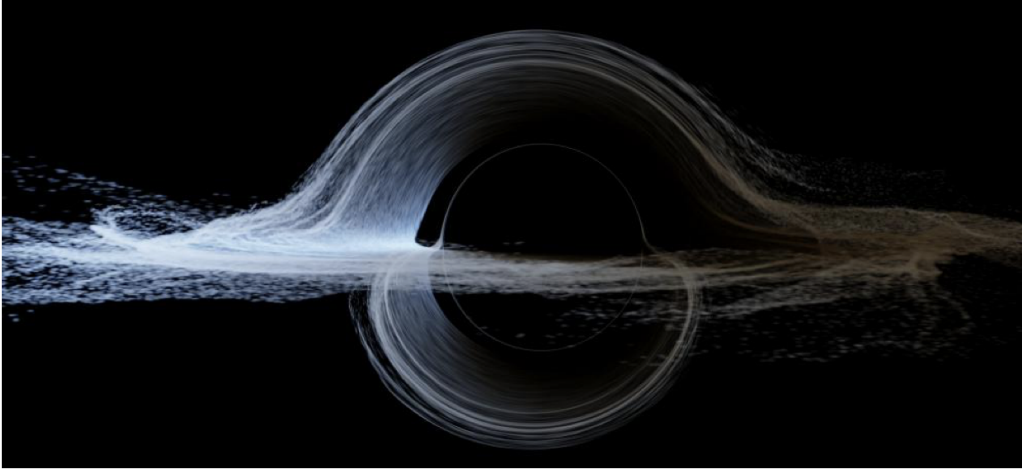


Figure 8.3: Image taken from Thorne’s paper simulating a realistic Doppler effect.

This image shows the expected Doppler effect for the accretion disk with their spin, $\frac{a}{M} = 0.6$. We can see the intensity and colour asymmetry that would actually occur. It is clear that the intensity of light emitted by matter moving away from the camera is significantly poor and this is why the producers opted for their own visual involving no colour or brightness shifts and their own added lens flare (figure 6.1).

Chapter 9

Conclusion

To conclude this dissertation, let us summarise what we have covered in this paper and talk about further areas on research in this field of study.

We began by introducing the concept of spacetime and in turn general relativity. Laying out the conceptual foundation of spacetime and curvature allowed us to introduce the concept of the spacetime interval and how we apply this to define distances in space. We looked at the simplest spacetime interval, the distance measurement in flat space. Upon challenging ourselves as to how we could measure distance in curved space, we established one of the most significant features in general relativity, the metric tensor. We concluded our first main chapter by giving examples of applying this metric.

The chapter detailing geodesics led us on the desired path of our focus. We further nurtured our understanding of the metric and curved spacetime by deriving the geodesic equations we would require for finding equations of motion in space under any given metric. Our derivation followed a mathematical physics approach utilising the action principle and Euler-Lagrange equations. Christoffel symbols were introduced and utilised in a simple example so that we truly grasped the concept beyond definition.

After establishing the geodesics we were able to move onto the focus of the dissertation, ray tracing under black holes. Before we could jump ahead to the main black hole of interest, the Kerr black hole, we needed to introduce the foundational black hole metric, the Schwarzschild metric. By focusing on this metric for a chapter we were able to bring all the concepts we studied into practice. We used the geodesic equations to develop equations of motion under this metric and also derived constants of motion for which defined the features of a Schwarzschild black hole whilst simplifying our equations of motion. This metric, being the foundational metric all other black hole metrics were built upon, laid out our basic understanding of a black hole's properties and equations of motion.

The Schwarzschild metric allowed us to develop simple ordinary differential equations which we require for ray tracing. We dedicated a chapter entirely to ray tracing under this metric where we produced not only 2D plots to boost our understanding, but also created our own lensed image of space containing a star field given the presence of a Schwarzschild black hole. We discussed what we would expect to see based on our derived properties and also introduced further features like accretion disks and gravitational redshift to challenge concepts required for a realistic black hole image. In developing our black hole visualisation we discussed the process of what we developed in Python step by step.

Having gained immense understanding of the ray tracing process under black hole metrics we were then able to move on to the final black hole metric, the Kerr metric. This metric is the metric used for the black hole Gargantua used in the movie *Interstellar* in 2014 and is our main topic of interest in this dissertation. In order to obtain the mathematical understanding of Gargantua we needed to take a similar approach as we did with the Schwarzschild metric. We defined the Kerr metric and sought after the equations of motion; however, the complexity of the Kerr metric allowed us to challenge our understanding of special relativity, specifically momenta.

We started by trying to find constants of motion first. General momenta can be derived through their metric components by applying what Einstein teaches us in special relativity, this gave us a fast way to derive two constants of motion. Instead of moving on to find the other constants, we sought after the link between generalised momenta and the geodesic equation to bridge our mathematical physics understanding with special relativity ensuring an in depth understanding of our methods. We utilised the Hamilton-Jacobi equation to derive the vital Carter constant from an ansatz, via additive separability. Having adopted four constants of motion we continued the use of the Hamilton-Jacobi equation to derive the uncoupled equations of motion with Mino time. The approach of using the Hamilton-Jacobi equation is linked closely with geodesics as we found with the momenta link, but ultimately saves a lot of gruelling calculations for us. We finished our most mathematically challenging chapter with deriving the features of the Kerr black hole as we did with the Schwarzschild black hole.

In the penultimate main chapter, we moved on to ray tracing under the Kerr metric where we followed very closely to the structure of ray tracing under the Schwarzschild metric. We developed our plots and discussed what we would expect to see given this metric. Furthermore, we talked about more advanced properties we would expect a detailed image of a Kerr black hole to have, adding the Doppler effect to the properties we discussed with the Schwarzschild black hole. However, the biggest concept we tackled was how we would adapt our process for developing our own visualisation in Python. We acknowledged the challenge asymmetry brought us for our visualisation production and introduced techniques utilised by Kip Thorne and the Double Negative team for *Interstellar*

To end the dissertation, the final main chapter brought us to Gargantua, the Kerr black hole. We delved into the DNGR (the code used to produce Gargantua) coding process in a familiar way to our own visualisations. This approach allowed us to challenge our own methods without going into excessive detail that would otherwise be found in Thorne's paper. The biggest difference we saw was how they overcame the challenge of producing an image from a close distance to the black hole as opposed to our distant one. We linked back to the expectations of what we would expect to see in a detailed visualisation and criticised Gargantua based on its accuracy. Our critique discussed the purposeful down play and omission of certain effects due to the purpose of the visualisation in the movie, which was to ultimately look visually stunning and serve the plot.

Overall, we were able to confidently walk our way through the ray tracing process of two significant black holes types utilising many mathematical methods to develop the equations of motion needed. Our conceptual understanding of black holes should either be solidified or improved along with our understanding of light trajectories through curved spacetime. This concept can be applied in improving our models for image generation and even expanding

into other black hole metrics which account for features like charge. Ray tracing can also be applied to worm holes which remains relevant to the topic of Interstellar.

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